

R002 — Figure atlas: multiple shooting with guess propagation for sparse model selection

Concept figures, fresh simulations supporting the theory, solver/optimiser/parameter variations,
and the repository defects that were found and fixed

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report folder `reports/2026-09-11-R002-multishooting-figure-atlas/`

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Part I (extended landscape animations) added 12 September 2026

Abstract

This atlas collects, in presentation order, every figure produced for the `hiddenSelect` project (Julia; multiple shooting with guess propagation for discovering sparse polynomial governing equations from noisy trajectories; theory manuscript *Sparse Optimization using Multi-Shooting and Guess Propagation*). It contains: (i) seven concept figures that replace the manuscript’s hand-drawn sketches (the inverse problem, single shooting, the shooting partition, multiple shooting, node removal, the guess-propagation algorithm, and the error-growth picture); (ii) numerical checks of Lemma 1, Lemma 2 and Proposition 1 of the manuscript on the FitzHugh–Nagumo (FHN) system; (iii) cost-landscape figures showing how the number of local minima, the Hessian spread and the blow-up region grow with the window size; (iv) the headline experiment — guess propagation versus restarting from the seed — over 8 optimiser seeds with paired bootstrap confidence intervals, plus variations of the window-size schedule, blow-up penalty, noise level, sparsity weight, optimiser, iteration budget, integrator sub-steps and start distance; (v) an integrator/solver comparison; (vi) the quantification of the defects reported in the code review R001 (F1, F2, F4, F5) and of two *new* defects found here (a mixed monomial ordering in the current driver and a global-variable shadowing in the single-shooting loss), all fixed in the live tree with this report; (vii) the same method on Lotka–Volterra and Lorenz. Every number in every caption is computed by scripts in this folder from a frozen code snapshot and asserted by 20 gates (§13); twenty MP4 animations (including a screened set of five FHN cost-landscape planes at 201×201 and four rotating 3-D landscapes, §12) and a slide deck are shipped alongside. Headline: with guess propagation the median parameter error over seeds falls from 1.65 (seed) to 0.32 at full single shooting, while the same optimiser without propagation stays at the seed (1.65) — paired difference -1.14 , 95 % bootstrap CI $[-1.43, -0.77]$, better in 7 of 8 seeds (the eighth seed starts on the blow-up plateau, where nothing can move; a graded penalty fixes that).

Table of figures

Figure	What is plotted	What it demonstrates / how to use it on a slide
<i>A. Concept figures (replace the hand-drawn sketches of the manuscript)</i>		
1	FHN trajectory, noisy samples, phase portrait, the 2×10 sparse coefficient matrix	The inverse problem: recover the few non-zero library coefficients from noisy samples. Opening slide.
2	Single-shooting model trajectory from y_0 at a wrong p , residual lines, $ \text{residual} $ vs t	Why single shooting is hard: the initial-condition error is amplified like e^{Lt} over the whole record.
3	Timeline of data times t_i , shooting nodes τ_k , windows, Δt , ΔT ; the coarser partition below	The shooting partition and what “removing a node” means (replaces <code>shooting_partition.png</code>).
4	Single vs multiple shooting ($\kappa = 100, 10, 5$) at the same wrong p , per-window segments, residuals	Restarting at the nodes caps the error at $e^{L\Delta T}$ (replaces <code>multiple_shooting.png</code>).

Figure	What is plotted	What it demonstrates / how to use it on a slide
5	Fine vs coarse partition around a removed node, ΔT_1 , ΔT_2 , the bound of Prop. 1	Node removal: what changes in the cost when one node is dropped (replaces <code>comparison_shooting_nodes.png</code>).
6	1-D cost slices for $\kappa = 1, 5, 25, 100$ with the minimiser carried forward; pseudo-code	The guess-propagation algorithm: solve the smooth problem first, carry its minimiser into the rugged ones.
7	Schematic error envelopes $e^{L\Delta T}\ \eta\ $ vs the saw-tooth capped at $e^{L\Delta T}\ \eta\ $	The one-line argument for multiple shooting (uses the report's L and $\ \eta\ _{\max}$).
<i>B. Numerical checks of the theory</i>		
8	$\ \varphi(t; x_1) - \varphi(t; x_2)\ /\ x_1 - x_2\ $ for 24 probes vs $e^{\pm Lt}$, $e^{\mu t}$	Lemma 1 holds; the Lipschitz bound is very loose on the FHN limit cycle (actual amplification ≤ 8).
9	$\ \varphi(t; p_1) - \varphi(t; p_2)\ /\ p_1 - p_2\ $ vs the Lemma 2 bound	Lemma 2 holds; parameter sensitivity saturates instead of growing exponentially.
10	$ \hat{J}_K - J_K $ vs number of removed nodes and vs ΔT_2	Proposition 1 qualitatively: the cost change grows with $ I_R $ and with ΔT_2 ; remove few nodes at a time.
<i>C. Cost landscapes</i>		
11	2-D slices of J_κ in the (v, v^3) coefficient plane of $\dot{v}$, $\kappa = 1, 3, 10, 100$	The basin narrows and a blow-up plateau appears as κ grows; a positive cubic coefficient blows up in finite time.
12	Same in the $(w$ in $\dot{v}$, v in $\dot{w})$ plane	The slow $\dot{w}$ equation is weakly identifiable at small κ and better at large κ .
13	1-D slices along random directions; local-minima count vs κ	Local minima per slice grow from 1.3 ($\kappa = 1$) to 7.9 ($\kappa = 100$).
14	Hessian $\lambda_{\max}$, $ \lambda_{\min} $, $\ \nabla J_\kappa(p^*)\ $, number of negative eigenvalues vs κ ; sensitivity to x_0	Conditioning worsens by $10^{4.5}$ and the noisy cost becomes indefinite at p^* for $\kappa \geq 2$.
<i>D. Guess propagation: the headline experiment (8 seeds)</i>		
15	J_κ , $\ p - p^*\ $ and recovery score vs κ for GP, best-guess GP and no propagation	The claim of the paper, with seeds shown: GP reaches single shooting with $\ p - p^*\ \approx 0.32$; the control never leaves the seed.
16	Paired differences GP – control with bootstrap CIs; cellwise counts	The difference is decisive at every $\kappa \geq 2$ (7 of 8 seeds; CI excludes 0).
17	Seed-median coefficient error heat-map along the sweep	Which coefficients are recovered when; the w coefficient of $\dot{v}$ is shrunk by the sparsity term.
18	Filmstrip of single-shooting fits from the minimiser at $\kappa = 1, 5, 25, 100$, GP vs control (seed 2)	Visual companion of 15; still frames of <code>anim_sweep.gif</code> .
19	Nelder–Mead cost traces per κ (seed 2), GP vs control	GP starts every stage near its optimum; the control stalls on the plateau.
20	Final coefficients at $\kappa = 100$ vs truth, all seeds	Correctly labelled coefficient bar chart (the repository's labels were stale).
<i>E. Variations</i>		
21	Window-size schedule: dense (44 steps), full (16), coarse (4), jump (2)	Removing few nodes per step matters: jump 0.76 vs dense 0.30 median error.
22	Flat vs graded blow-up penalty	The graded penalty removes the “stuck on the plateau” failure (0 vs 12.5 % of seeds).
23	Noise level $\sigma_{\text{rel}} \in \{0, \dots, 0.2\}$	GP is robust up to 10 % noise and fails at 20 %; the control fails even at zero noise.
24	Sparsity weight γ	$\gamma = 0.05$ is best; $\gamma = 0$ and $\gamma = 1$ both hurt; which coefficients survive.
25	Nelder–Mead vs BFGS vs L-BFGS (AD gradient through the windowed loss)	With the F4 fix, BFGS+GP is the best and cheapest arm ($\ p - p^*\ = 0.21$, score 0.95); gradients die on the plateau.
26	Basin of attraction of one window size vs start distance	Small windows have wide basins but a noise-limited floor; single shooting has a narrow basin.
27	Nelder–Mead iteration budget; integrator sub-steps S	2500 iterations are needed; $S \geq 5$ suffices; $S = 1$ biases the fit.

Figure	What is plotted	What it demonstrates / how to use it on a slide
28	Start scale 0.01–0.3, flat vs graded penalty	Farther starts are harder; the graded penalty rescues starts that begin on the plateau.
<i>F. Integrators and solvers inside the loss</i>		
29	Accuracy vs cost per loss evaluation for 14 integrator settings	The in-house fixed-step Tsit5 ($S = 10$) matches a 10^{-12} reference to 10^{-13} and is the cheapest accurate choice; the problem is not stiff.
30	GP sweep with three integrators inside the loss	The integrator does not change the outcome; the plateau start (seed 1) does.
<i>G. Defects found and fixed</i>		
31	F2: share of the loss dropped by the mod 7 stiff branch; 1-D slice through the $\dot{w}$ equation	9–17% of $J(p^*)$ and up to 78% away from p^* was silently discarded; fixed.
32	F1: Jacobian entry, eigenvalues and trace from the truncated vs fixed RHS	The stability movies used $\partial\dot{v}/\partial v = +1.0$ instead of -1.25 ; fixed.
33	F4: cosine between the gradient the sweep received and the right one; BFGS with each	Near-orthogonal at $\kappa = 1$; BFGS stalls with the wrong gradient; fixed.
34	New: mixed monomial order in the mod 8 driver; global-data shadowing	A spurious $\ p - p^*\ = 2.50$ and an ignored argument; both fixed.
<i>H. Other systems and closing the loop</i>		
35	Lotka–Volterra (12 coefficients)	GP recovers the system ($\ p - p^*\ = 0.29$, score 0.83); the control blows up in 5 of 6 seeds.
36	Lorenz (30 coefficients, chaotic)	GP+BFGS recovers Lorenz (score 0.83); GP+Nelder–Mead does not converge; the control fails in 4 of 4 seeds.
37	Hessian at the GP minimisers; cost along the segment between consecutive minimisers; cross-evaluation	The strong-convexity premise is not verified at Nelder–Mead’s returned iterates (positive definite in 36% of cells); the carried guess sits inside the next basin.

Animations (MP4, H.264; frames regenerable by `figures/make_animations.py` and, for the last two groups, `figures/make_landscape_animations.py`; the GIF `anim_sweep.gif` is the same content as the first one). The first nine are 825×616 ; the extended landscape set of §12 is 2560×1440 :

- `figures/anim_fhn_sweep.mp4` — FHN: $v(t)$ fit from the minimiser at every stage, GP vs control (seed 2), with the running cost and error.
- `figures/anim_fhn_phase.mp4` — FHN: the same sweep in the (v, w) phase plane with the coefficient bars against the truth.
- `figures/anim_fhn_multiple_shooting.mp4` — concept: single shooting drawn forward in time at a wrong p , then the $\kappa = 10$ windows appearing one by one, then node removal ($\kappa = 1 \rightarrow 100$) with the cost of the wrong and true p .
- `figures/anim_fhn_landscape.mp4` — FHN: the 2-D cost landscape in the (v, v^3) plane morphing through every stage of the FULL schedule, with the GP and control paths (seed 2) overlaid (data: `anim_landscape_v_v3.csv`, script `analysis/10_animation_data.jl`).
- `figures/anim_lv_sweep.mp4`, `figures/anim_lv_phase.mp4` — Lotka–Volterra: $x(t), y(t)$ fits, phase plane and coefficients along the sweep, GP vs control (seed 1).
- `figures/anim_lv_landscape.mp4` — Lotka–Volterra: the 2-D cost landscape in the (x, xy) coefficient plane of $\dot{x}$ morphing through every FULL window size, with the GP and control paths (seed 1) overlaid (data: `anim_lv_landscape_x_xy.csv`, script `analysis/11_lv_landscape.jl`).
- `figures/anim_lv_landscape_x2_xy.mp4`, `figures/anim_lv_landscape_xy_xy.mp4` — the same at 161×161 resolution in two more planes: (x^2, xy) of $\dot{x}$, where the true-zero quadratic term produces finite-time blow-up (the analogue of the FHN cubic: a smooth diagonal valley at $\kappa = 1$ shatters into a thin sliver with secondary minima at $\kappa = 100$), and the interaction pair $(xy$ in $\dot{x}$, xy in $\dot{y})$, where a round bowl becomes an L-shaped plateau (data: `anim_lv_landscape_x2_xy.csv`, `anim_lv_landscape_xy_xy.csv`).
- **Extended 2-D landscapes** (§12; 201×201 grids, 1440p; data `anim_fhn_landscape_*.csv.gz`, `anim_lv_landscape_x2_xy_hires.csv.gz`, script `analysis/12_landscape_planes.jl`) —

`anim_fhn_landscape_v_v3_hires.mp4` (the Part C plane at the new resolution), `.._wv_ww_hires.mp4` (v and w in $\dot{w}$: the slow block, nearly flat at $\kappa = 1$), `.._v2_w2_hires.mp4` (v^2 in $\dot{v}$ and in $\dot{w}$: two true zeros, plateau $0 \rightarrow 57\%$), `.._w_ww_hires.mp4` (w in $\dot{v}$, v in $\dot{w}$: the feedback pair; sharpens hard but barely blows up — plateau 0% for every $\kappa \leq 33$, 0.27% at $\kappa = 100$), `.._v3_w3_hires.mp4` (v^3 in $\dot{v}$ and in $\dot{w}$: rugged at both ends), and `anim_lv_landscape_x2_xy_hires.mp4`. Each frame also carries the basin and plateau fractions of that κ .

- **3-D landscapes** (§12; rotating surface $z = \log_{10} J_\kappa$, one full turn over 288 frames) — `anim3d_fhn_landscape_wv_ww.mp4`, `anim3d_fhn_landscape_v2_w2.mp4`, `anim3d_fhn_landscape_v_v3.mp4` and `anim3d_lv_landscape_x2_xy.mp4`. The surface is clipped at the 99.5th percentile of the finite costs for display; see the display convention in §12.
- **3-D landscape without guess propagation** — `anim3d_fhn_landscape_wv_ww_control.mp4`: the same slow-block surface as `anim3d_fhn_landscape_wv_ww.mp4` with the guess-propagation path removed, showing only the minimiser the optimiser returns at each κ when every stage restarts from the same seed — what the method does *before* guess propagation is introduced. The gutter panel carries that arm's $\|p^{(\kappa)} - p^*\|$ along the ladder.

A slide deck telling the story in 8 slides is in `slides/multishooting_story.pptx` (built by `slides/make_deck.py` from these figures and `tables/table_numbers.json`). Every figure exists as PDF and PNG under `figures/`.

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1 Introduction

The code-review report R001 (10 September 2026) established that the repository’s central claim reproduces and listed seven defects. This report is the figure companion for a presentation. It adds three things R001 did not have: proper concept figures for the method (Part A), a broad batch of *fresh* simulations run in parallel (522 optimisation sweeps over 8 seeds and 9 variation axes, plus two other dynamical systems) that support or qualify the theory’s claims (Parts B–F, H), and the quantification *and fix* of the defects, with before/after figures (Part G). Nothing is re-used from R001: every quantity that overlaps (the headline sweep, the F1/F2 numbers) is recomputed here from a frozen snapshot of the sources, and the bug figures execute the pre-fix and post-fix code side by side.

Two things the reader should keep in mind. First, the optimiser seed matters enormously and is reported everywhere: one of the eight random starts lands on the blow-up plateau at the very first window size and never moves under the repository’s flat penalty (§7); it is kept in every statistic rather than dropped. Second, the sweep in the committed driver (`fhn_model_selection_mod 8.jl`) had guess propagation *disabled* and used two different monomial orderings for the sweep and the polish (§10); the driver is fixed in the same commit as this report.

2 Definitions (read before the figures)

Everything used in a caption is defined here, including terms inherited from the manuscript and from R001.

2.1 Model, library and data

The target is FitzHugh–Nagumo,

$$\dot{v} = v - \frac{1}{3}v^3 - w + 0.5, \quad \dot{w} = \frac{1}{12.5} (v + 0.7 - 0.8w), \quad (1)$$

written as a general two-dimensional cubic polynomial system so that model *selection* (which monomials are present) is part of the task:

$$\dot{x}_i = \sum_{j=1}^J p_{ij} \Theta_j(\mathbf{x}), \quad i = 1, 2, \quad \Theta = \{1, w, w^2, w^3, v, vw, vw^2, v^2, v^2w, v^3\}, \quad J = 10. \quad (2)$$

Read this: the library order above is the *multi-index* order produced by `helper_functions.jl` and used by `odefun_poly!` (mod 8). The older hard-coded RHS `odefun` (mods 2–7) uses the order

$1, v, w, v^2, vw, w^2, v^3, v^2w, vw^2, w^3$. The two orders are different permutations of the same 10 monomials; mixing them is finding N1 in §10. All coefficient plots here use the library order, with the $\dot{v}$ block (10 entries) followed by the $\dot{w}$ block. The parameter vector $p \in \mathbb{R}^{20}$ has 6 non-zero entries: $\dot{v}$: (1)=0.5, (w)=-1, (v)=1, (v^3)=-1/3; $\dot{w}$: (1)=0.056, (w)=-0.064, (v)=0.08 (seven including the $\dot{w}$ constant). The optimisation vector is $z = [x_0; p] \in \mathbb{R}^{22}$: the first two entries are a free initial condition that enters only through the term $\|x_0 - y_0\|^2$.

Data follow the repository recipe exactly (frozen mod 8 lines 103–157, reproduced to 6×10^{-15} , gate G3): integrate with the in-house fixed-step Tsit5 at $\delta t_{\text{fine}} = 0.01$ from (1, 1) to $T = 156$, keep every 100th point, crop to $t \in [50, 150]$ and shift to $t \in [0, 100]$ ($N = 101$ samples, spacing $\Delta t = 1$), and add Gaussian noise to each component with standard deviation $0.05 \times$ that component’s own standard deviation (seed 1287436679). The resulting maximum noise norm is $\|\eta\|_{\text{max}} = 0.159$ and the noise floor of the data term is $\frac{1}{N} \sum_i \|\eta_i\|^2 = 0.0052$.

2.2 Cost function

For a window size κ (number of data intervals per window, $1 \leq \kappa \leq N-1$) the record is cut into $K = \lceil (N-1)/\kappa \rceil$ windows; each window is integrated *from the noisy datum at its left node* with S fixed Tsit5 sub-steps of length $\delta t = \Delta t/S$ per data interval, and

$$J_\kappa(z) = \frac{1}{N} \left[\|x_0 - y_0\|^2 + \sum_{i=1}^{N-1} \|\hat{x}(t_{i+1}; p, y_{\tau(i)}) - y_{i+1}\|^2 \right] + \frac{\gamma}{20} \sum_{j=1}^{20} \text{smooth } \ell_1(p_j), \quad (3)$$

where $\tau(i)$ is the last node at or before t_i , and $\text{smooth } \ell_1(x) = \alpha^{-1} [\log(1 + e^{-\alpha x}) + \log(1 + e^{\alpha x})]$ with $\alpha = 500$ (equal to $|x|$ for $|\alpha x| > 40$). $\kappa = 1$ is one window per data interval; $\kappa = N - 1 = 100$ is single shooting. **Read this:** node states are *not* optimisation variables and no continuity constraint is imposed (unlike textbook multiple shooting); this is the manuscript’s formulation and the repository’s code. If any state exceeds 10^3 in magnitude or becomes NaN during integration, the loss returns the constant 10^3 (*flat blow-up penalty*, a plateau with zero gradient). The *graded* penalty (a variant introduced here) returns $10^3 (1 + \frac{N-1-i}{N-1}) + J_{\text{partial}}/N$ when the blow-up occurs at data index i , so that later blow-ups cost less and a derivative-free optimiser has a direction. The manuscript’s J_K (no normalisation, no sparsity term) differs from (3) by the factor $1/N$ and the γ term; the theory figures (Part B) use $\gamma = 0$.

2.3 Arms

GP (guess propagation) Sweep κ upward through a schedule; the minimiser of each stage is the start of the next. Schedules: FULL = {1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 25, 33, 50, 75, 100} (default); SHORT = {1, 2, 5, 10, 25, 50, 100} for the variation experiments; DENSE = {1, ..., 30, 35, 40, ..., 100} (44 stages); COARSE = {1, 5, 25, 100}; JUMP = {1, 100}.

best-GP As GP but the start of the next stage is the best (lowest-cost) minimiser seen so far.

control (no propagation) Every stage starts from the same random seed vector. This is the arm the committed mod 8 driver was running.

optimiser Nelder–Mead (Optim.jl, 2500 iterations, default tolerances) unless stated; BFGS and L-BFGS use the exact ForwardDiff gradient of (3) through the integrator (possible only after fix F4), same iteration cap, $g_{\text{tol}} = 10^{-8}$.

seed Optimiser seed $s \in \{1, \dots, 8\}$: $z_{\text{seed}} = [y_0; 0.1\xi]$, $\xi \sim \mathcal{N}(0, I_{20})$ from `MersenneTwister(s)`; $\|p_0 - p^*\| \in [1.50, 1.84]$ over the 8 seeds (0.1 is R001’s scale; the repository uses 0.01 — see 28).

Read this: arms are compared at *equal* budgets (same iterations, same schedule, same seeds) except where the figure says otherwise: the schedule experiment (21) deliberately changes the number of stages, and the optimiser experiment (25) reports wall time because the gradient methods do different work per iteration.

2.4 Metrics

J_κ (**primary**) the cost (3) at the returned minimiser, including the sparsity term; $J_\kappa \geq 10^3$ flags a blow-up (*plateau*). Costs at different κ are not directly comparable (different restarts).

$\|p - p^*\|$ Euclidean distance between the fitted 20-vector and the true coefficients; the identification metric. Not computed by the repository; the same definition as R001.

recovery score fractional system-recovery score in $[0, 1]$: each of the 20 coefficients earns credit 1 if it is correctly zero ($|p_j| < 0.05$ when $p_j^* = 0$) or correctly non-zero within 25 % of the true value; the score is the mean credit. *support recall* (fraction of the 7 true non-zeros found) and *false positives* (true zeros with $|p_j| \geq 0.05$) are its components.

paired difference for two arms sharing a seed, the per-seed difference of $\|p - p^*\|$ at a given κ ; reported as the mean with a 95 % percentile bootstrap CI over 2000 resamples of the *seeds* (resampling unit = optimiser seed). “Cellwise” counts report in how many seeds one arm is strictly better.

wall time seconds on one core for one stage; only used comparatively and never in a gate.

2.5 Symbols and frozen constants

symbol	meaning	value	where frozen
N	number of samples	101	data recipe (mod 8)
Δt	sample spacing	1.0	data recipe
$S, \delta t$	integrator sub-steps per interval, step	10, 0.1	driver (27 varies S)
κ, K	window size (data intervals), number of windows	swept, $\lceil 100/\kappa \rceil$	schedules above
γ, α	sparsity weight, smooth- ℓ_1 sharpness	5×10^{-2} , 500	driver (24 varies γ)
σ_{rel}	relative noise per component	0.05	data recipe (23 varies)
$\ \eta\ _{\text{max}}$	largest noise norm in the data	0.159	<code>fhn_data_meta.json</code>
L	Lipschitz constant $\sup \ \partial f / \partial x\ _2$ along the true orbit / on the box $[-2.5, 2.5] \times [-1, 2]$	3.05 / 5.34	<code>concept_meta</code>
μ	logarithmic norm $\sup \lambda_{\text{max}}((J + J^T)/2)$ along the orbit	1.17	idem
$\tilde{L}$	$\sup \ \partial f / \partial p\ _2$ along the orbit (Lemma 2)	10.56	idem
$\Delta T, \Delta T_1, \Delta T_2$	window length $\kappa \Delta t$; fine window length; $\tau_k - \tau_k^-$ (Prop. 1)	—	manuscript
blow-up threshold	$ x > 10^3$ or NaN	10^3	loss

Read this: L is measured, not assumed. Along the FHN limit cycle the actual amplification of an initial-state perturbation never exceeds ≈ 8 (8), so every bound that carries $e^{L\Delta T}$ is loose by orders of magnitude for $\Delta T \gtrsim 2$; the bounds are qualitatively right (they order the arms correctly) but not quantitatively tight.

3 Provenance and what was executed

All numerics run from the frozen snapshot `deps/` (post-fix sources) through `analysis/common.jl` and `analysis/hs_core.jl`, a corrected, allocation-light, dimension-generic re-implementation of the windowed loss that is gated against the frozen mod 8 `forward_simulation_loss_windows` (relative difference 1.9×10^{-13} over 42 evaluations, gate G2) and against the frozen mod 8 data block (6×10^{-15} , G3). The bug scripts additionally execute `deps/prefix/` (pre-fix sources, byte-identical to commit 44abf4a, `origin/main` at freezing time — `git fetch` was run first). SHA-256 of both snapshots: `deps/MANIFEST.md` (G1). Julia 1.11.3 with the committed environment and no package added; figures by matplotlib from CSV/JSON; nothing in Julia plots. `Distributed` cannot open sockets in the sandbox, so the 522 sweep jobs ran as 12+8+6 independent sharded processes (`REPRODUCE.md`). Realised coverage (every axis value and cell count, computed by `analysis/make_inventory.py`) is in `planning/coverage_matrix.md`; result-file hashes in `analysis/results/results_hashes.json`; every number quoted in prose is listed with its source in `analysis/number_manifest.md`.

4 Part A — Concept figures

These are the slide figures. They use the real FHN data of §2 and a deliberately *wrong* parameter vector p_{wrong} (cubic coefficient -0.30 instead of $-1/3$, $\dot{w}$ time-scale multiplied by 1.15; $\|p_{\text{wrong}} - p^*\| = 0.038$) so that the

a

The inverse problem: recover the few non-zero entries of p from noisy samples of v, w

-- true trajectory • v samples y_i • w samples y_i

one data interval $\Delta t = 1$

$N + 1 = 101$ samples, $\Delta t = 1$, 5 % relative noise

state

time t

b

grey arrows: vector field of the true FHN model

-- true orbit
• noisy samples

Data-generating model (FitzHugh–Nagumo)

$$\dot{v} = v - v^3 / 3 - w + 0.5$$

$$\dot{w} = (v + 0.7 - 0.8 w) / 12.5$$

Ansatz $\dot{x}_i = \sum_j p_{ij} \Theta_j(v, w)$, where

$\Theta =$ every monomial in (v, w) of degree ≤ 3
(10 terms per equation $\Rightarrow$ 20 unknowns p_{ij}).

c

	1	w	w ²	w ³	v	vw	vw ²	v ²	v ² w	v ³
$\dot{v}$	0.5	-1			1					-0.333
$\dot{w}$	0.056	-0.064			0.08					

unknown sparse coefficients p (20 entries, 7 non-zero)

library monomial $\Theta_j(v, w)$

blank cell = true zero; cell colour = sign of p_{ij} , shaded by $\sqrt{|p_{ij}|}$

8

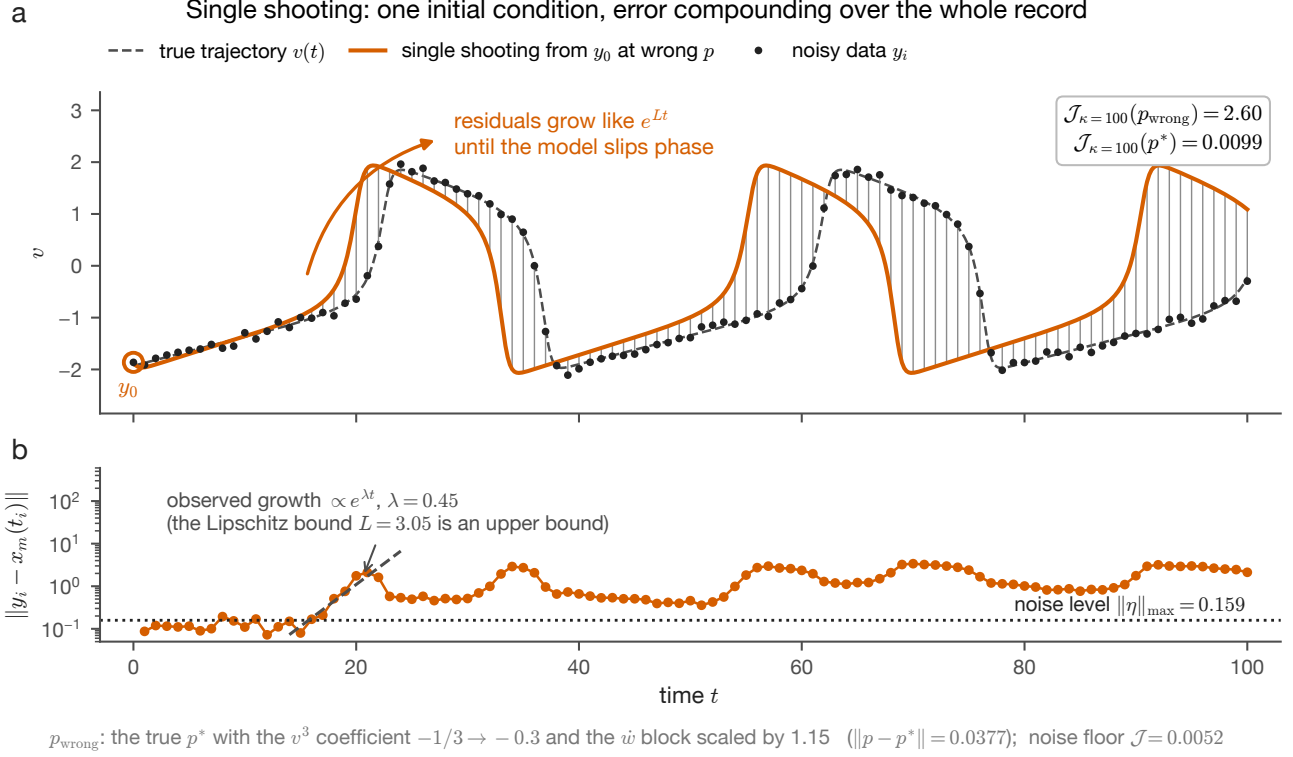
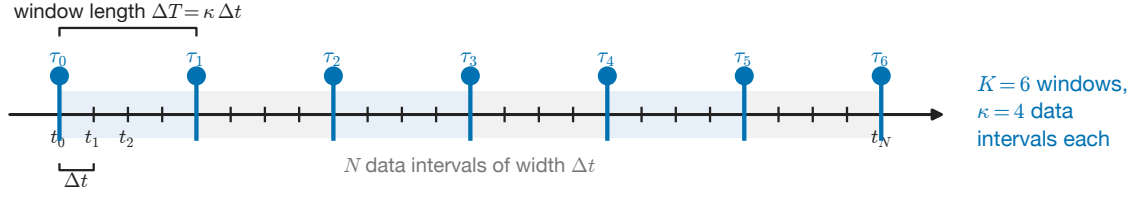


Figure 2: Single shooting. The model is integrated once from the first datum y_0 over the whole record with the wrong parameter p_{wrong} (vermilion); thin vertical lines are the residuals to the data. The residual (right, log scale) sits at the noise level until the model slips phase near $t \approx 18$, then grows at an observed rate ≈ 0.45 per time unit (the Lipschitz bound $L = 3.05$ is an upper bound, far above it) and saturates at the size of the attractor; the cost is $\mathcal{J}_{100}(p_{\text{wrong}}) = 2.60$ (vs 0.0099 at p^*), so a tiny parameter error produces a large, non-convex cost. Data: `concept_segments.csv` (`param=wrong`, `window_size=100`), `concept_costs.csv`.

^a Fine partition: a shooting node τ_k at every κ -th datum; each window restarts from the datum $y_{\tau_{k-1}}$



^b Node removal: delete every other node $\Rightarrow$ half as many windows, each twice as long

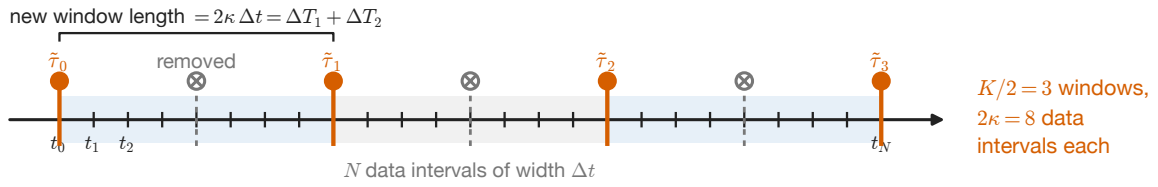


Figure 3: The shooting partition. Top: the data times $t_0, \dots, t_N$ (ticks, spacing Δt) and the shooting nodes $\tau_0 < \tau_1 < \dots < \tau_K$ placed at every κ -th datum; the windows $[\tau_{k-1}, \tau_k]$ of length $\Delta T = \kappa \Delta t$ are shaded alternately. Bottom: the coarser partition obtained by removing every other node ($K/2$ windows of length $2\Delta T$) — the elementary step of guess propagation. Schematic (no data); replaces the manuscript's `shooting_partition.png`.

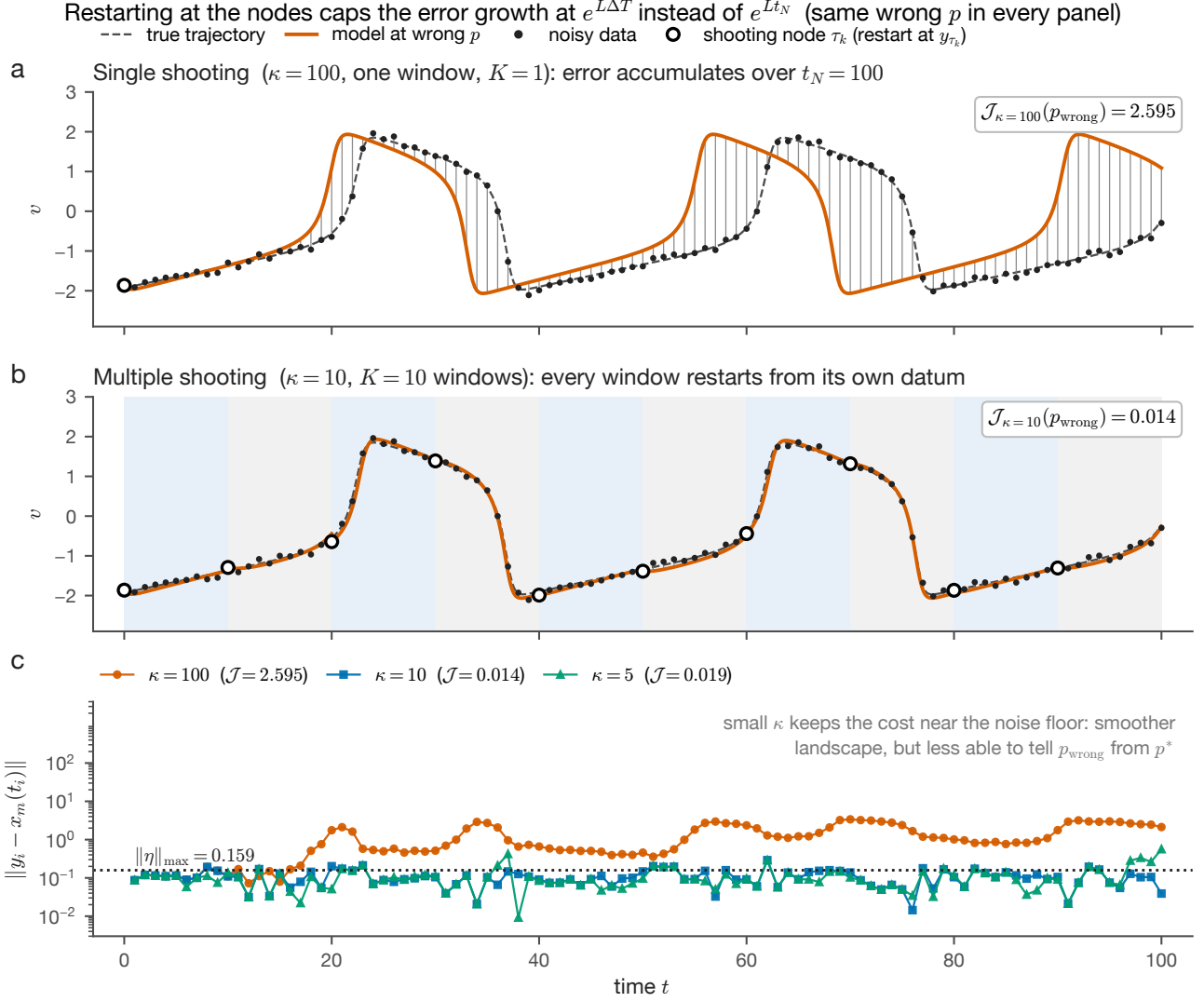
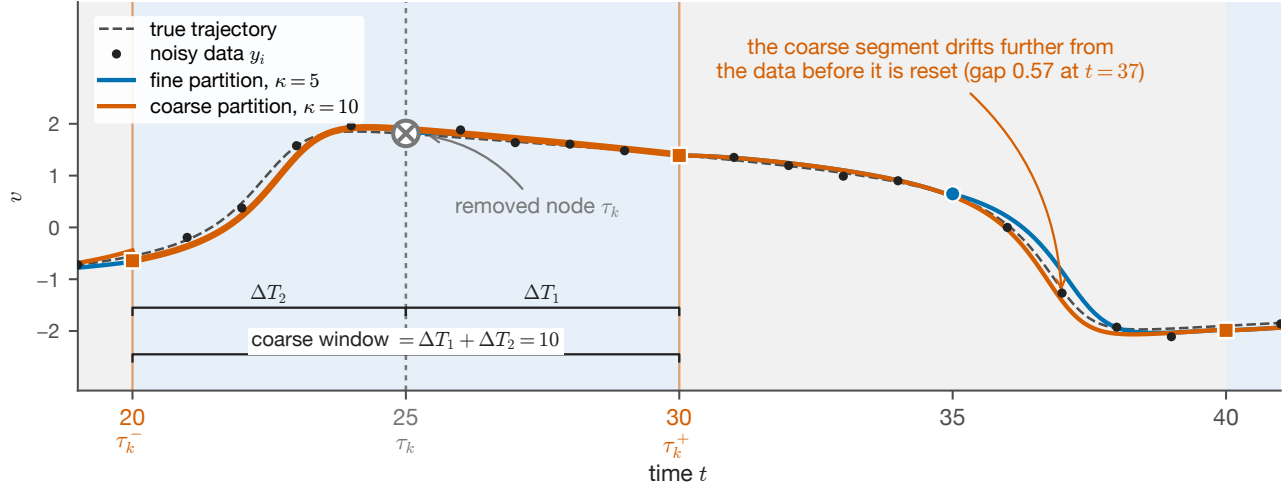


Figure 4: Multiple shooting. The same wrong parameter as 2, integrated (a) as single shooting ($\kappa = 100$, $J = 2.60$) and (b) restarting from the datum at every 10th sample ($\kappa = 10$, $J = 0.0140$); windows are shaded, node markers highlighted, residual lines drawn to the data; (c) the per-datum residual magnitude on a log scale for $\kappa = 100, 10, 5$ ($J = 2.60, 0.0140, 0.0188$). Restarting resets the accumulated error at every node, so the exponential amplification is paid over $\Delta T = \kappa \Delta t$ instead of over the whole record; with $\kappa \leq 10$ the wrong model is only 1.4–1.8 $\times$ the cost of the true one, with $\kappa = 100$ it is 260 $\times$. The flip side, visible in (c): short windows keep even a wrong model near the noise floor, i.e. they give a smooth landscape at the price of less power to discriminate p from p^* — which is why the sweep must end at single shooting. Replaces `multiple_shooting.png`. Data: `concept_segments.csv`, `concept_costs.csv`.

Removing the node τ_k : the segment launched from τ_k^- must now cover $[\tau_k^-, \tau_k^+]$



$$\|\hat{\mathcal{J}}_K(p) - \mathcal{J}_K(p)\| \leq 2 |I_R| C_{\max} \left(e^{L(\Delta T_1 + \Delta T_2)} \|\eta\|_{\max} + e^{L\Delta T_1} \left[\frac{\tilde{L}}{L} (e^{L\Delta T_2} - 1) \|p - p^*\| + \|\eta\|_{\max} \right] \right)$$

$\Delta T_1 = \max_k(\tau_k - \tau_{k-1})$ = fine window length; $\Delta T_2 = \max_k(\tau_k - \tau_k^-)$ = distance from the removed node to its left neighbour (both = 5 here); $|I_R|$ = number of removed nodes; $C_{\max}$ = largest segment/data magnitude over the affected windows $L = 3.05$, $\tilde{L} = 10.56$ (Lipschitz in x , in p); $\|\eta\|_{\max} = 0.159$. Background shading: the coarse ($\kappa = 10$) windows.

Figure 5: Node removal (Proposition 1). Zoom on $t \in [19, 41]$ with the wrong parameter: the fine partition ($\kappa = 5$, blue segments) and the coarse one after removing the node τ_k at $t = 25$ (open marker; $\kappa = 10$, vermilion), with $\tau_k^- = 20$ and $\tau_k^+ = 30$; the measured gap between the fine and coarse segments at $t = 37$ is printed. After removal the segment launched at the left neighbour τ_k^- must cover $[\tau_k^-, \tau_k^+]$; the cost changes only on the windows adjacent to the removed nodes, by at most $2|I_R| C_{\max} \left(e^{L(\Delta T_1 + \Delta T_2)} \|\eta\|_{\max} + e^{L\Delta T_1} \left[\frac{\tilde{L}}{L} (e^{L\Delta T_2} - 1) \|p - p^*\| + \|\eta\|_{\max} \right] \right)$ with ΔT_1 the fine window length and $\Delta T_2 = \tau_k - \tau_k^-$. Design rule: remove few nodes at a time, prefer low-noise nodes, keep ΔT_2 small. Replaces `comparison_shooting_nodes.png`.

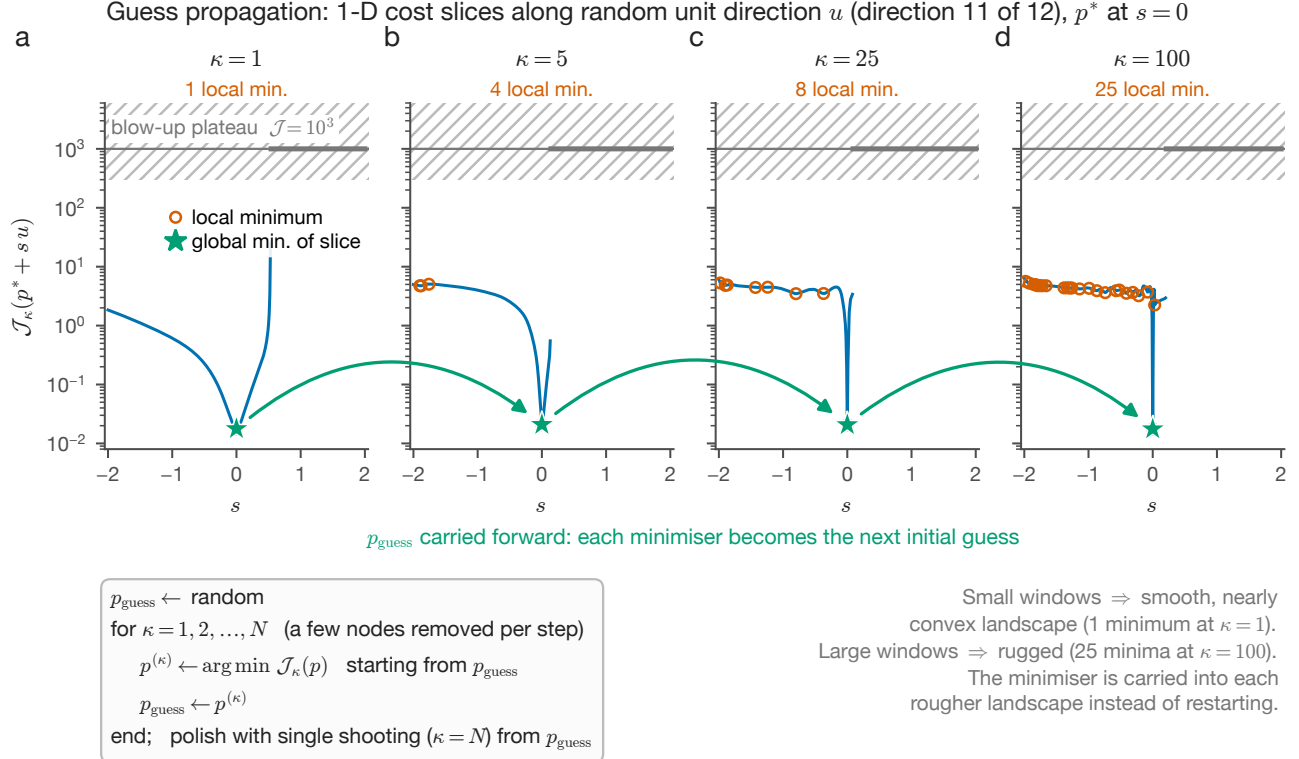


Figure 6: The guess-propagation algorithm. One-dimensional slices $J_\kappa(p^* + s u)$ along the same random unit direction u (the one of the 12 directions of 13 with the most local minima at $\kappa = 100$: 1, 4, 8, 25 local minima at $\kappa = 1, 5, 25, 100$, circled) for $\kappa = 1, 5, 25, 100$ (log scale; hatched band = blow-up plateau $J = 10^3$; star = global minimum, which is at $s = 0$ because the slices pass through p^*). Small windows give a smooth, nearly convex landscape; large windows add local minima and a wide plateau. The algorithm solves the $\kappa = 1$ problem from a random start and carries each minimiser (arrow) into the next, more rugged landscape, ending with a single-shooting polish. Data: `landscape_1d_slices.csv`.

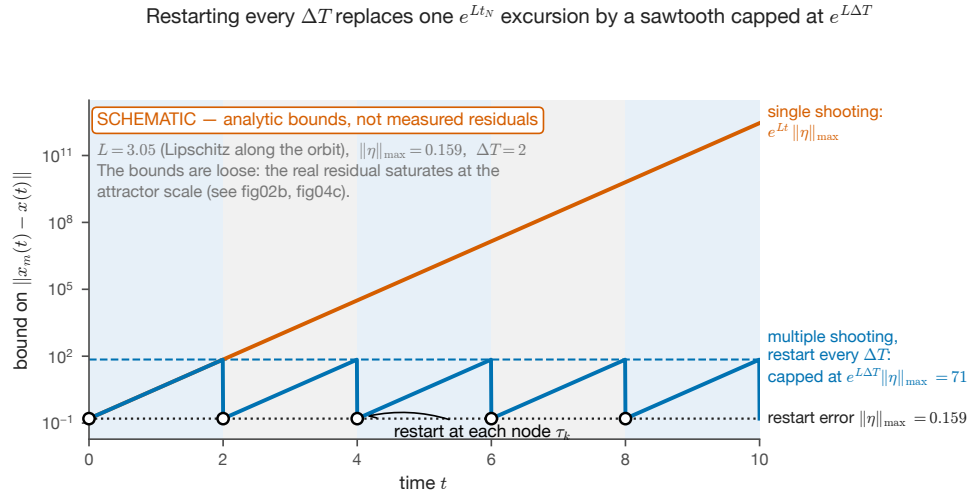


Figure 7: Error growth, schematic. The Lemma 1 envelope $e^{L t} \|\eta\|_{\max}$ for single shooting (one restart at t_0) versus the sawtooth of multiple shooting, capped at $e^{L \Delta T} \|\eta\|_{\max}$ by the restart at every node, drawn with the report's $L = 3.05$ and $\|\eta\|_{\max} = 0.159$. This is the manuscript's one-line argument; 8 shows how loose the envelope is on the actual system.

5 Part B — Numerical checks of the theory

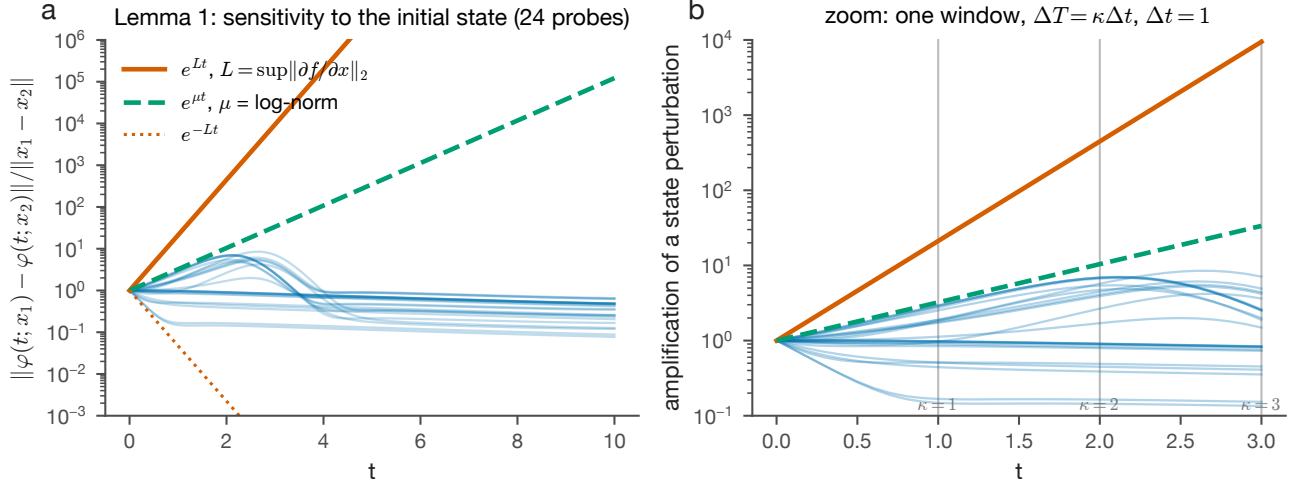


Figure 8: Lemma 1 (flow sensitivity to the initial state). Amplification $\|\varphi(t; p^*, x_1) - \varphi(t; p^*, x_2)\| / \|x_1 - x_2\|$ for 24 probes (x_1 = a datum at 4 different phases of the orbit, $x_2 = x_1 + 10^{-3}u$ for 6 random unit directions; blue), against e^{Lt} with $L = \sup_{\text{orbit}} \|\partial f / \partial x\|_2 = 3.05$ (vermilion), e^{-Lt} (dotted) and $e^{\mu t}$ with the logarithmic norm $\mu = 1.17$ (green dashed). (a) 10 time units, (b) zoom on the first three windows. The two-sided bound holds everywhere, but the actual amplification peaks at ≈ 8 near the fast jump of the relaxation oscillation and then *contracts* (the limit cycle is attracting), whereas e^{Lt} reaches 10^{13} at $t = 10$. Data: lemma1_flow_sensitivity.csv, concept_meta.json.

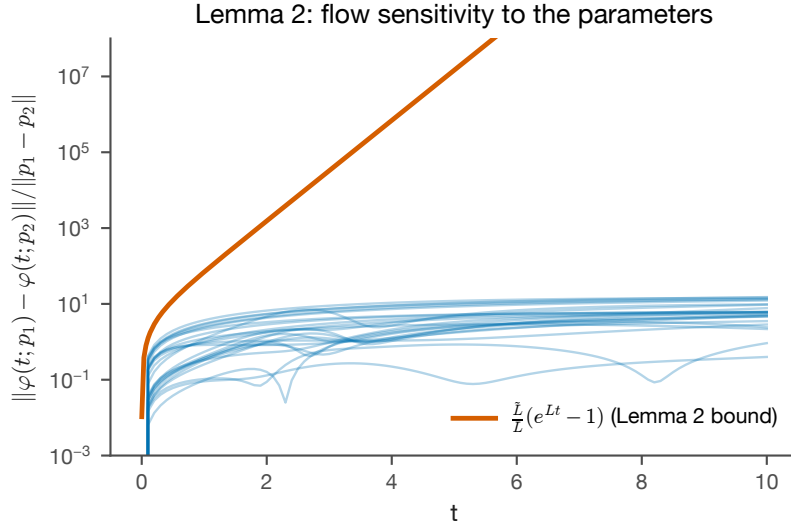


Figure 9: Lemma 2 (flow sensitivity to the parameters). $\|\varphi(t; p_1, x_0) - \varphi(t; p_2, x_0)\| / \|p_1 - p_2\|$ for 24 probes ($p_2 = p^* + 10^{-3}u$, random unit $u \in \mathbb{R}^{20}$) versus the bound $\frac{\tilde{L}}{L}(e^{Lt} - 1)$ with $\tilde{L} = \sup_{\text{orbit}} \|\partial f / \partial p\|_2 = 10.56$. The bound holds; the measured sensitivity saturates around 10^1 – 10^2 because the orbit is attracting. Data: lemma2_param_sensitivity.csv.

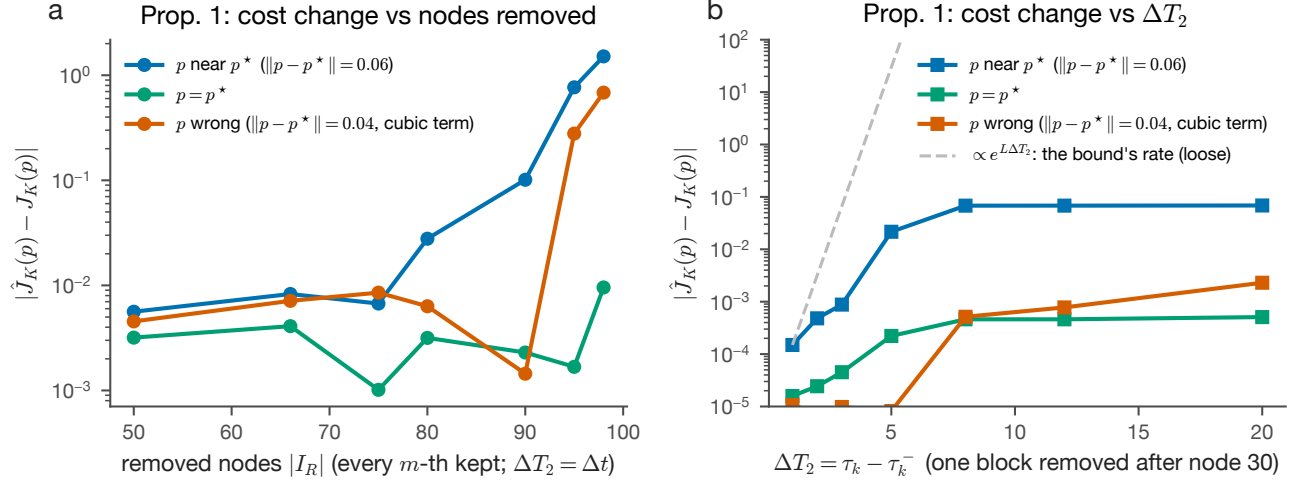


Figure 10: Proposition 1 (node removal changes the cost by a bounded amount). Starting from the finest partition (every datum a node, $K = 100$) with $\gamma = 0$: (a) keep every m -th node so that $|I_R| = 100 - \lceil 100/m \rceil$ nodes are removed ($\Delta T_2 = \Delta t$ fixed, $\Delta T_1 = m\Delta t$ grows), and (b) remove one contiguous block of b nodes after node 30 ($|I_R| = b$, $\Delta T_2 = b\Delta t$ grows). Shown for $p = p^*$ (green), a vector 0.02 away from p^* on 6 coefficients (blue, $\|p - p^*\| = 0.06$) and p_{wrong} of Part A (vermillion, $\|p - p^*\| = 0.04$ concentrated on the cubic term). $|\hat{J} - J|$ grows with both $|I_R|$ and ΔT_2 , and much faster for the wrong parameter, as the proposition says; the bound's rate $e^{L\Delta T_2}$ (grey dashed in (b)) is far steeper than the observed growth, which saturates once the removed block spans a full fast excursion. Data: `prop1_node_removal.csv`.

6 Part C — Cost landscapes

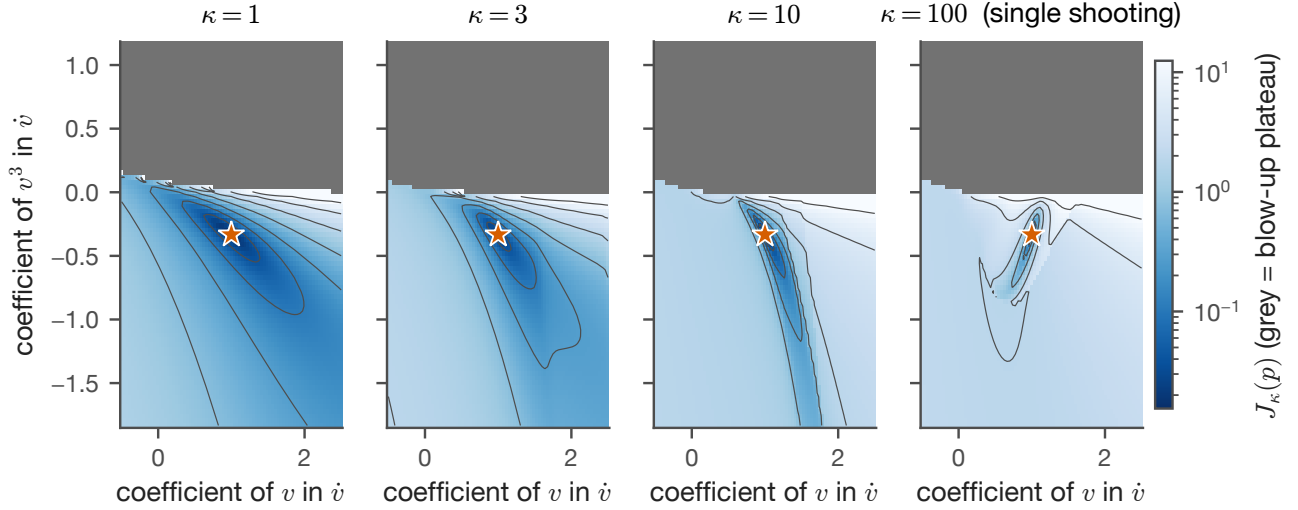


Figure 11: 2-D slices of J_κ in the (v, v^3) coefficient plane of the $\dot{v}$ equation. All other coefficients at their true values, $x_0 = y_0$, $\gamma = 0.05$; star = truth $(1, -1/3)$; grey = blow-up plateau ($J \geq 10^3$); colour is $\log J$. As κ grows from 1 to 100 the basin around the truth narrows by an order of magnitude and the plateau expands: any positive cubic coefficient (above the horizontal edge) makes $\dot{v} \sim cv^3$ blow up in finite time, and at $\kappa = 100$ even v -coefficients > 1.6 do. This is the geometric reason the control arm fails at large κ . Data: `landscape_2d_v_v3.csv`.

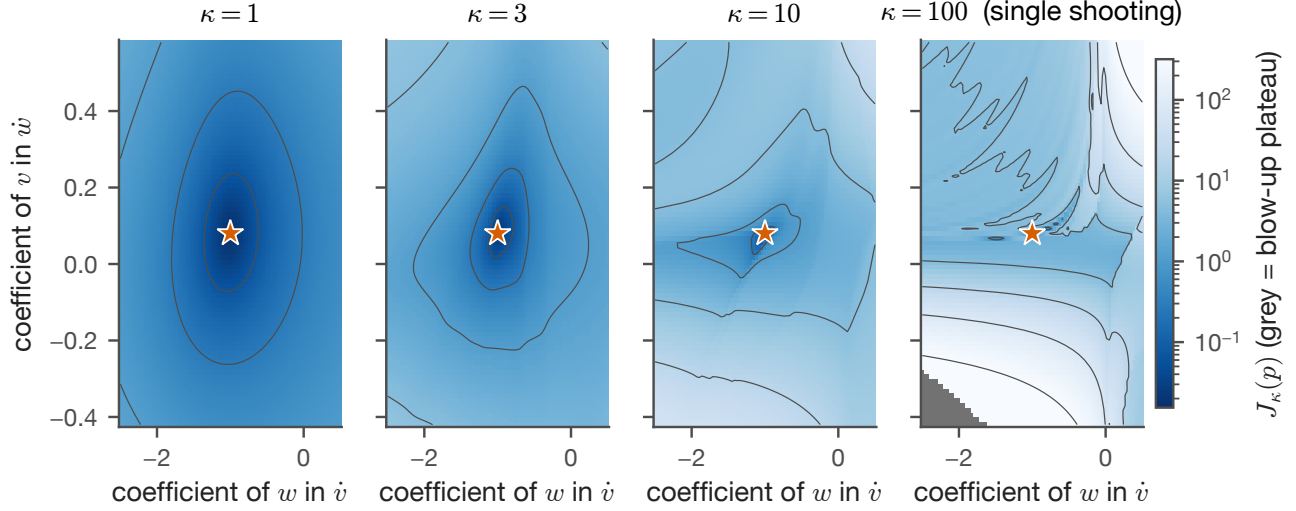


Figure 12: 2-D slices in the $(w \text{ in } \dot{v}, v \text{ in } \dot{w})$ plane. Same protocol as 11; star = truth $(-1, 0.08)$. The $\dot{w}$ coefficient (vertical axis, true value 0.08) is a slow time-scale: at $\kappa = 1$ the cost is almost flat along it (weakly identifiable from one-interval windows), and only long windows ($\kappa \geq 10$) pin it down — the mechanism behind the non-monotone parameter error along the sweep (15b). Data: `landscape_2d_w_wv.csv`.

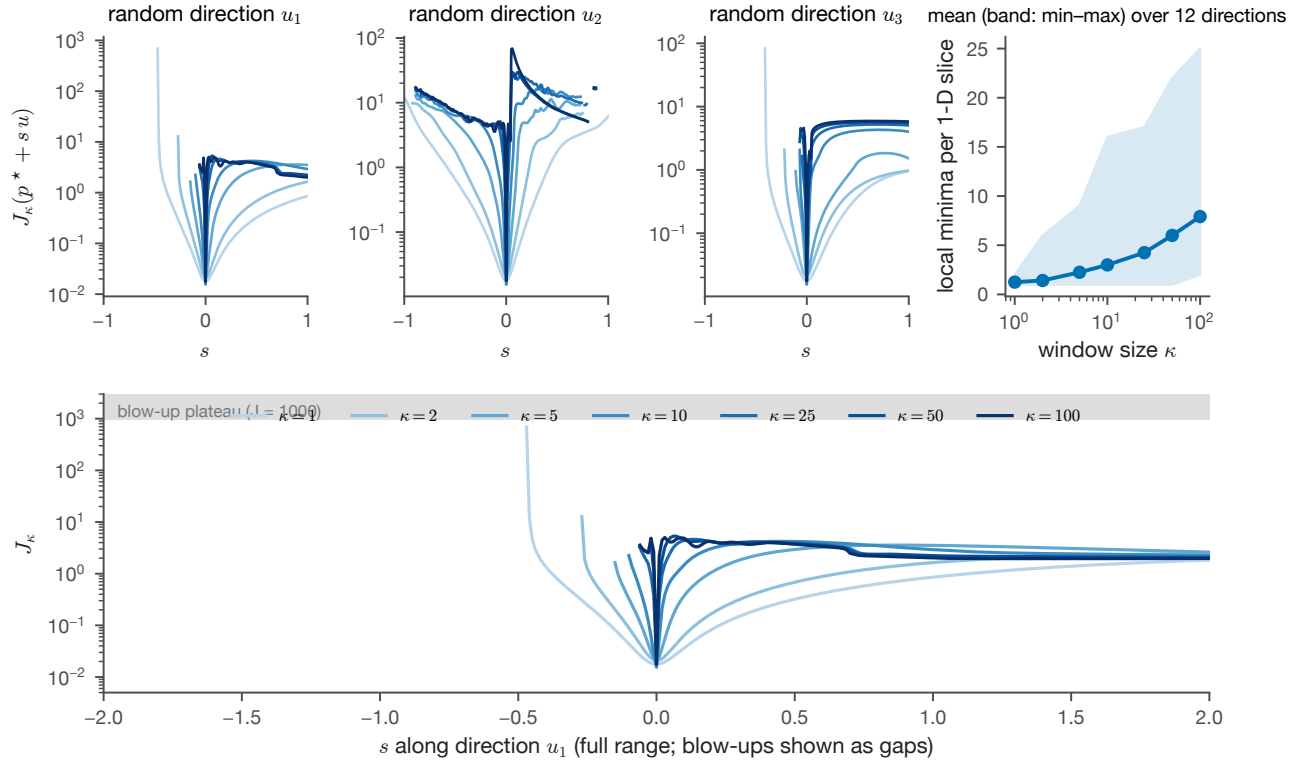


Figure 13: 1-D slices along random directions and the number of local minima. Top row: $J_\kappa(p^* + s u_d)$ for three random unit directions $u_d \in \mathbb{R}^{20}$ (colour light to dark = $\kappa = 1, 2, 5, 10, 25, 50, 100$; gaps = plateau); top right: local minima per slice (mean over 12 directions, band = min-max) rising from 1.25 at $\kappa = 1$ to 7.9 at $\kappa = 100$. Bottom: the full $s \in [-2, 2]$ range for direction 1. Larger windows sharpen the minimum (better conditioning at the truth) but add spurious minima and cut the finite region down to $|s| \lesssim 0.5$. Data: `landscape_1d_slices.csv`, `landscape_1d_minima.csv`.

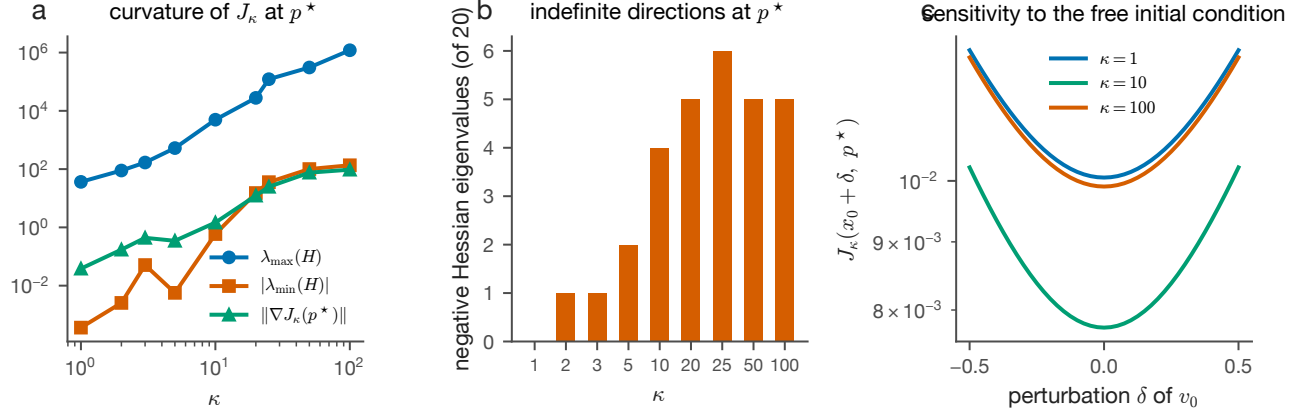


Figure 14: Curvature of the data term at p^* versus window size. (a) Largest and smallest Hessian eigenvalues of the data term of J_κ at p^* (ForwardDiff, 20 parameters) and the gradient norm: $\lambda_{\max}$ grows from 36 ($\kappa = 1$) to 1.2×10^6 ($\kappa = 100$) while $\|\nabla J_\kappa(p^*)\|$ grows from 0.04 to 96 — with noisy data the truth is not a stationary point and drifts away from the minimiser as κ grows. (b) The number of negative eigenvalues: the noisy cost is indefinite at p^* for every $\kappa \geq 2$ (up to 6 of 20 directions), so the manuscript’s strong-convexity assumption can only hold in a neighbourhood of the *minimiser* $p^{(\kappa)}$, not of p^* (checked there in 37). (c) Sensitivity of J_κ to the free initial-condition variable v_0 : a shallow parabola for every κ , because x_0 only enters through the first residual. Data: `hessian_at_ptrue.csv`, `landscape_x0_sensitivity.csv`.

7 Part D — Guess propagation: the headline experiment

Protocol: 8 optimiser seeds $\times$ 3 arms $\times$ the 16-stage FULL schedule, Nelder–Mead with 2500 iterations per stage, $\gamma = 0.05$, $S = 10$, flat penalty. Table 2 gives the seed medians and the paired statistics per stage.

Table 2: Headline sweep, seed medians over 8 seeds (GP = guess propagation, best-GP = best-guess propagation, control = no propagation) and the paired difference GP – control in $\|p - p^*\|$ with its 95 % bootstrap CI; “#GP better” counts seeds where GP is strictly better; “#GP on plateau” counts GP seeds whose minimiser is a blow-up (at $\kappa = 1$ three seeds are on the plateau initially; two escape at $\kappa = 2$ because the plateau at $\kappa = 1$ is not flat for their simplex — see text). Generated by `tables/make_tables.py` from `sweeps.csv`.

κ	J GP	J best-GP	J control	$\ p - p^*\ $ GP	$\ p - p^*\ $ best-GP	$\ p - p^*\ $ control	score GP	score control	GP–control	CI lo	CI hi	#GP better	#GP on plateau
1	0.02569	0.02569	0.02569	0.8641	0.8641	0.8641	0.4	0.4	0	0	0	0	3
2	0.02081	0.02081	0.03726	0.3803	0.3803	1.314	0.75	0.275	-0.7724	-1.112	-0.4196	7	1
3	0.01864	0.019	0.05955	0.3748	0.3893	1.466	0.75	0.225	-0.8679	-1.309	-0.4865	7	1
4	0.01549	0.01577	0.09718	0.3325	0.3615	1.51	0.775	0.25	-0.9232	-1.248	-0.5652	7	1
5	0.01906	0.01889	0.1259	0.2218	0.221	1.543	0.775	0.2	-1.133	-1.654	-0.6627	7	1
6	0.02114	0.02054	0.2428	0.3441	0.4522	1.716	0.55	0.2	-1.159	-1.526	-0.7401	7	1
8	0.0161	0.0154	0.3185	0.3593	0.3964	1.59	0.675	0.275	-1.113	-1.36	-0.7559	7	1
10	0.01406	0.01438	0.3086	0.2995	0.3347	1.737	0.8	0.2	-1.234	-1.634	-0.7787	7	1
12	0.01556	0.0155	0.6561	0.278	0.3312	1.622	0.8	0.25	-1.163	-1.415	-0.7988	7	1
15	0.01414	0.01409	0.5398	0.3348	0.3402	1.574	0.725	0.275	-1.109	-1.327	-0.7739	7	1
20	0.01436	0.01459	0.5358	0.2425	0.2709	1.612	0.8	0.25	-1.236	-1.547	-0.8214	7	1
25	0.01345	0.01362	0.861	0.3113	0.3455	1.534	0.85	0.25	-1.111	-1.401	-0.7455	7	1
33	0.01375	0.01373	0.8233	0.3372	0.3758	1.498	0.825	0.225	-0.9927	-1.358	-0.5627	7	1
50	0.01327	0.01334	1.334	0.3353	0.3393	1.807	0.8	0.2	-1.63	-2.426	-0.9189	7	1
75	0.0128	0.01289	1.392	0.3223	0.3179	1.793	0.8	0.225	-1.167	-1.522	-0.7642	7	1
100	0.01273	0.01288	1.506	0.3216	0.3105	1.649	0.8	0.25	-1.138	-1.426	-0.7719	7	1

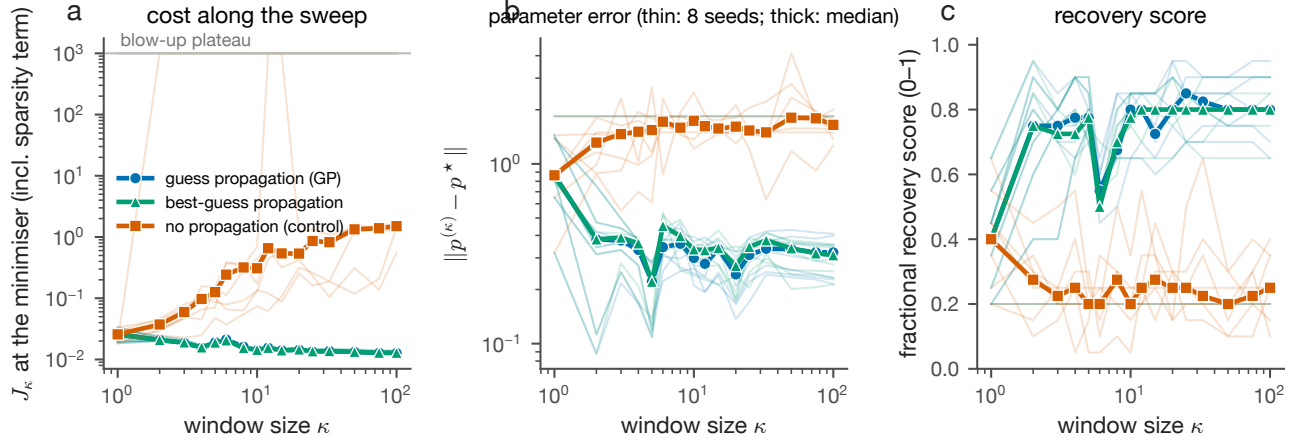


Figure 15: Guess propagation versus no propagation, 8 seeds. (a) Cost, (b) parameter error and (c) recovery score at the minimiser of each stage; thin lines = individual seeds, thick = median; blue circles GP, green triangles best-GP, vermilion squares control. All arms coincide at $\kappa = 1$ (same start). From $\kappa = 2$ on, GP drops to $\|p - p^*\| \approx 0.22\text{--}0.38$ and ends single shooting at 0.32 (score 0.80), whereas the control climbs back to the seed distance (1.65 at $\kappa = 100$; 25 % of its seeds end on the plateau, all others in spurious minima with J up to 1.5). One GP seed (seed 1) starts on the plateau at $\kappa = 1$ and never moves (flat line at 1.84). The parameter error is *not* monotone: it is smallest at $\kappa = 5$ (0.22) and drifts up to 0.32 while the cost keeps falling — the noise-induced bias of 14. Data: `sweeps.csv` (`exp=main`).

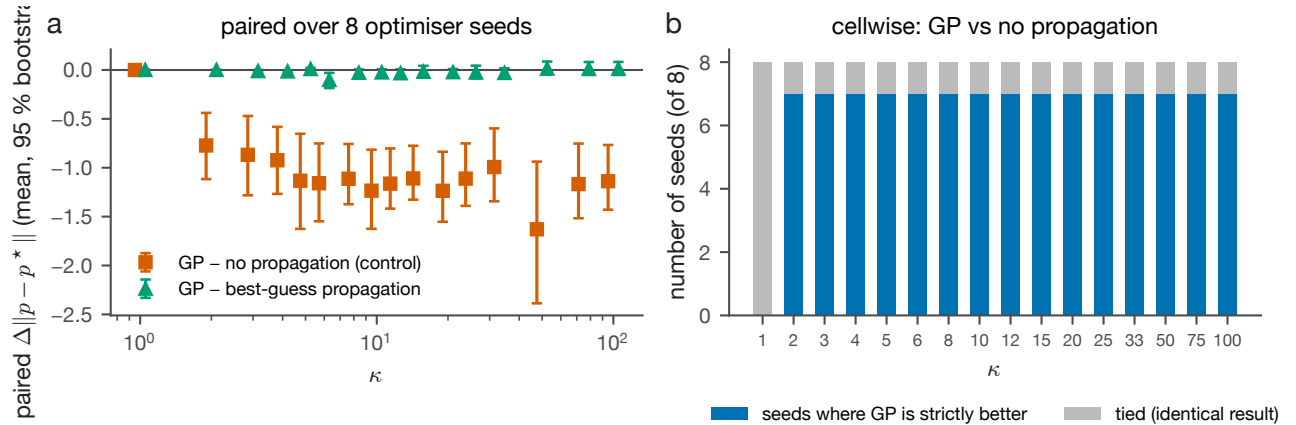


Figure 16: Paired comparison. (a) Mean paired difference in $\|p - p^*\|$ (GP minus control, vermilion squares; GP minus best-GP, green triangles) with 95 % bootstrap CIs over the 8 seeds; the GP-control difference is -0.77 at $\kappa = 2$, -1.14 $[-1.43, -0.77]$ at $\kappa = 100$, and its CI excludes zero at every $\kappa \geq 2$; GP and best-GP are statistically indistinguishable. (b) Cellwise counts: GP is strictly better than the control in 7 of 8 seeds at every $\kappa \geq 2$ and tied in the eighth (the plateau seed). Data: `derived_paired_main.csv`.

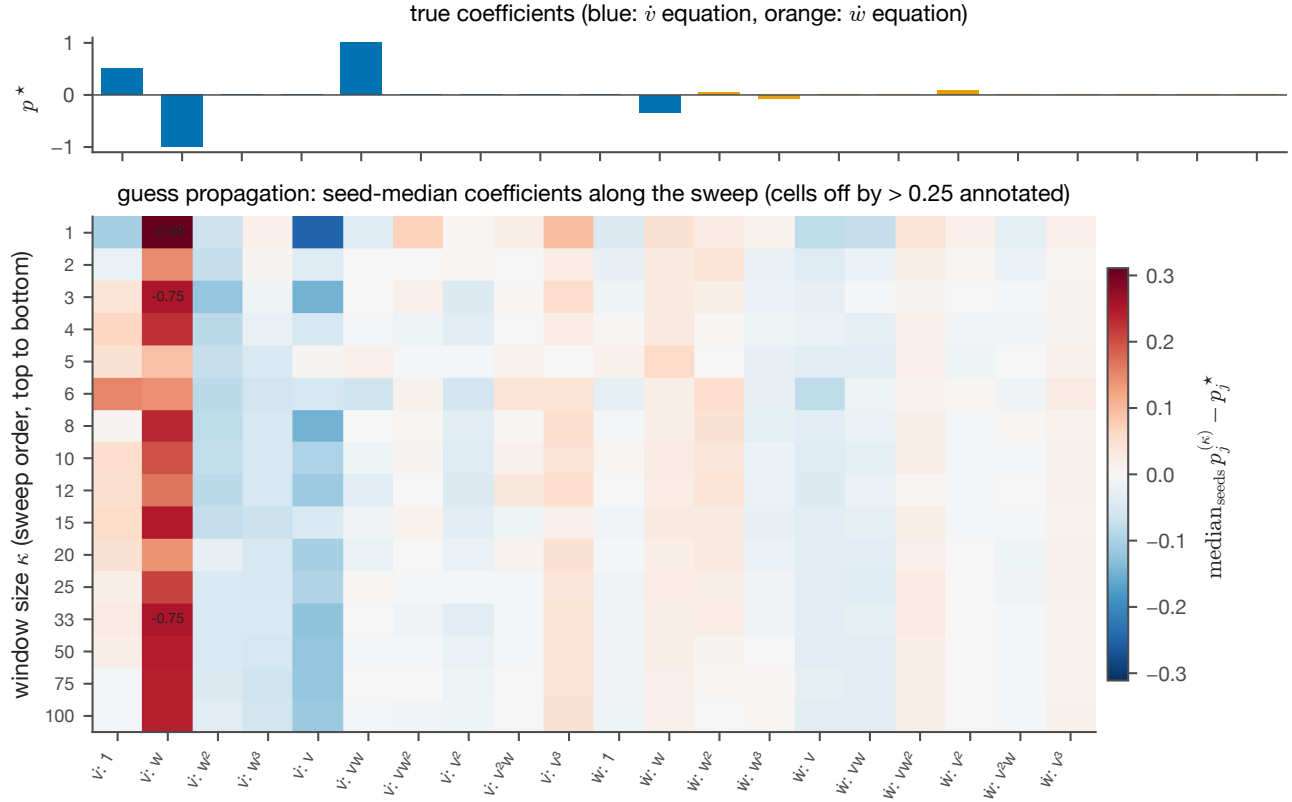


Figure 17: Which coefficients are recovered when. Top: the true coefficients (library order, $\dot{v}$ block blue, $\dot{w}$ block orange). Bottom: seed-median GP coefficient minus truth for every stage of the sweep (rows) and every coefficient (columns), diverging colour scale. The v and constant coefficients of $\dot{v}$ are right within 0.1 from $\kappa = 2$; the w coefficient of $\dot{v}$ stays ≈ 0.3 too small at every κ and the v coefficient ≈ 0.1 too small — the shrinkage of the smooth- ℓ_1 term at $\gamma = 0.05$ (cf. 24, where $\gamma = 0.01$ removes the shrinkage but loses sparsity); the slow $\dot{w}$ coefficients (≤ 0.08 in magnitude) are within 0.05. Data: `sweep_minimizers.csv`.

single-shooting simulation from the minimiser found at each window size (optimiser seed 2); grey: truth and data

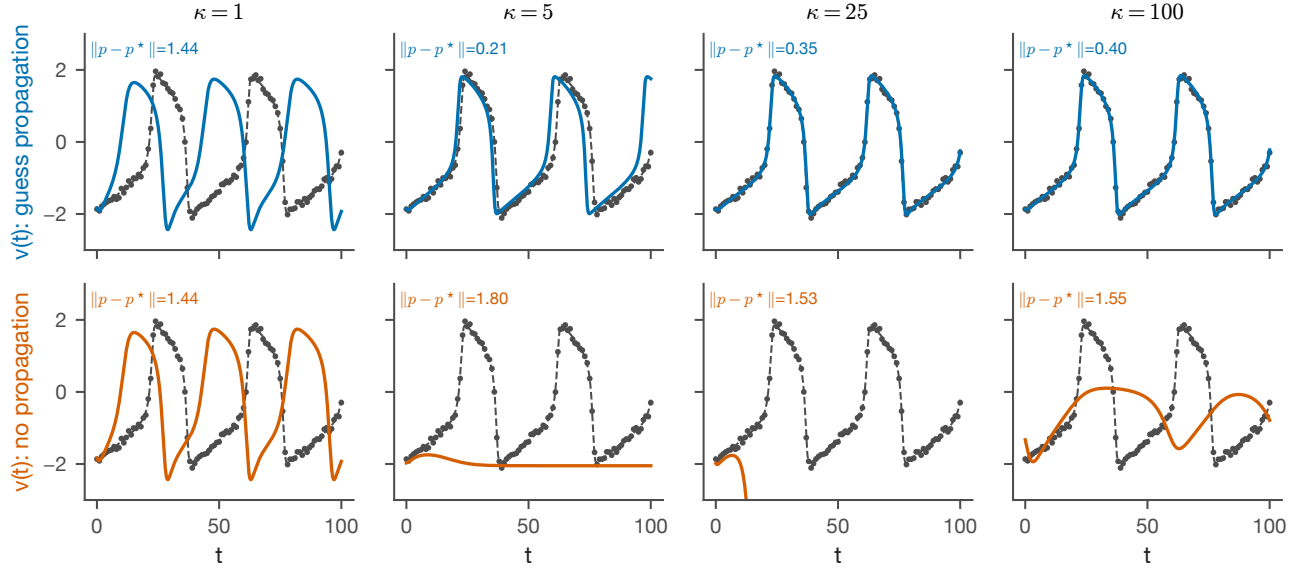


Figure 18: Filmstrip of the sweep (seed 2). Single-shooting simulation of $v(t)$ from the minimiser returned at $\kappa = 1, 5, 25, 100$ (columns) for GP (top, blue) and the control (bottom, vermilion); grey dashed = truth, dots = data; the parameter error is printed in each panel. The GP fit tracks the data at every stage; the control loses the trajectory as soon as the windows are long. Frames of `figures/anim_sweep.gif`. Data: `sweep_fits.csv`, `sweeps.csv`.

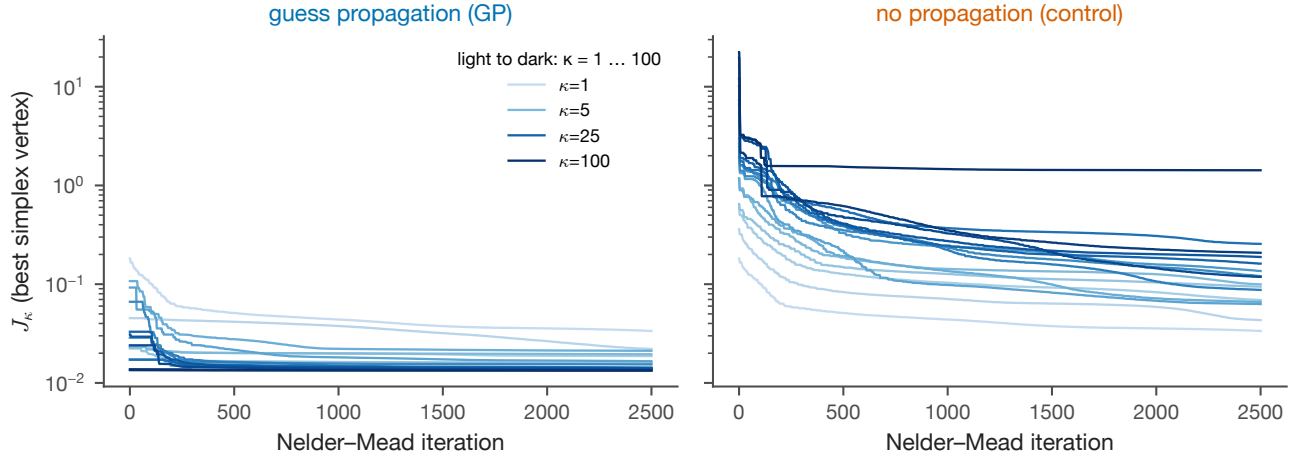


Figure 19: Optimiser traces (seed 2). Best simplex value of Nelder-Mead versus iteration for every stage (colour light to dark = $\kappa = 1 \dots 100$), GP (left) and control (right). With propagation each stage starts within a factor of a few of its final cost and converges; without it every stage starts from the same distant seed and, for $\kappa \geq 10$, from the plateau (flat traces at 10^3) or a spurious minimum. Data: `traces_seed2.csv`.

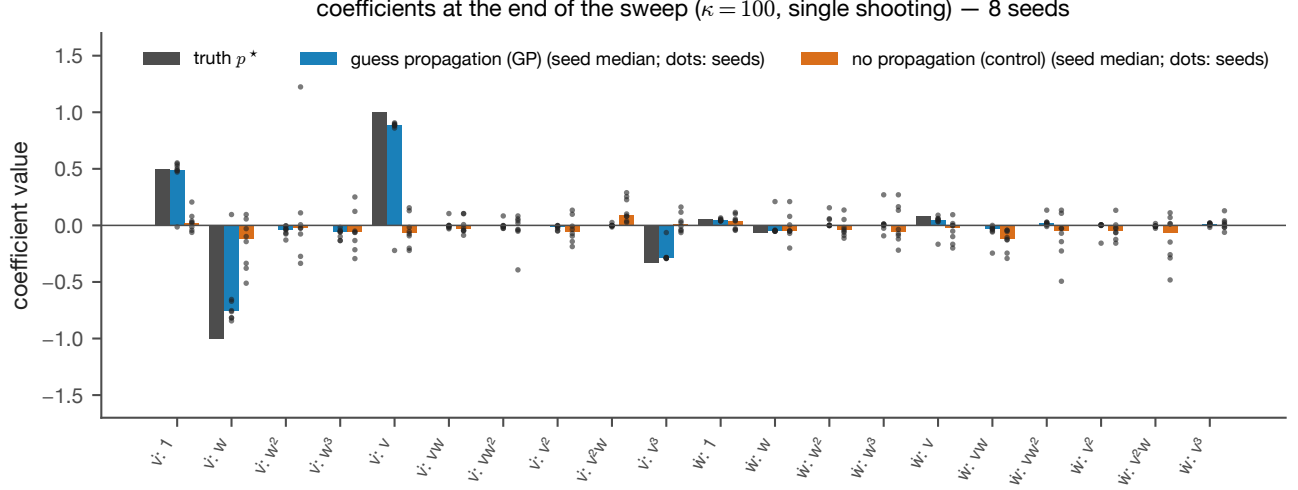


Figure 20: Coefficients at the end of the sweep ($\kappa = 100$). Truth (grey), GP seed-median (blue) and control seed-median (vermilion) with the 8 individual seeds as dots, correctly labelled in the library order (the repository’s bar-chart labels encoded the pre-refactor order — fix N1, §10). GP recovers the support (7 non-zeros, no false positive in the median) with the shrinkage noted in 17; the control’s median is near zero everywhere because its seeds scatter. Data: `sweep_minimizers.csv`.

Three GP seeds on the plateau at $\kappa = 1$, one forever. Table 2 shows 3 GP seeds with a blow-up minimiser at $\kappa = 1$ but only 1 from $\kappa = 2$ on. A random start with a positive cubic coefficient of order 0.1–0.3 blows up within one time unit ($\dot{v} \sim cv^3$, $|v| \approx 2$), so the loss is 10^3 at the start; if the whole initial Nelder–Mead simplex sits on the plateau the optimiser declares convergence after 0 iterations (all vertices equal) and returns the seed. Two of the three starts have a simplex vertex off the plateau at $\kappa = 2$ and escape; seed 1 does not, at any κ . This is a property of the *flat* penalty, not of guess propagation; the graded penalty (22, 28) removes it.

8 Part E — Variations

Table 3 lists the end-of-sweep ($\kappa = 100$) seed medians of every variation arm; each figure below shows the whole sweep.

Table 3: End-of-sweep ($\kappa = 100$) seed medians for every variation arm: $\|p - p^*\|$, recovery score, fraction of seeds on the plateau, and median wall time per stage. Arms within an experiment share seeds and schedule (SHORT for noise/sparsity/iters/substeps/seedscale, FULL for schedule/penalty/optimizer). “seedscale” arms are named `sc<scale>_<mode>_<penalty>`. Generated from `sweeps.csv`.

experiment	arm	n_seeds	$\ p - p^*\ $ ($\kappa=100$)	score	plateau frac	wall s per κ
schedule	coarse	8	0.5239	0.55	0.125	2.069
schedule	dense	8	0.3015	0.8	0.125	2.072
schedule	jump	8	0.7635	0.425	0.125	2.098
penalty	propagate_graded	8	0.274	0.825	0	2.107
penalty	reset_graded	8	1.631	0.3	0	1.88
noise	propagate $\sigma=0.0$	6	0.2313	0.8	0.1667	2.058
noise	propagate $\sigma=0.01$	6	0.3617	0.7	0.1667	2.117
noise	propagate $\sigma=0.02$	6	0.2482	0.875	0.1667	2.102
noise	propagate $\sigma=0.05$	6	0.2217	0.8	0.1667	2.087
noise	propagate $\sigma=0.1$	6	0.3135	0.625	0.1667	2.049
noise	propagate $\sigma=0.2$	6	1.881	0.225	0.1667	2.134
noise	reset $\sigma=0.0$	6	1.633	0.25	0.3333	1.845
noise	reset $\sigma=0.01$	6	1.602	0.2	0.3333	1.878
noise	reset $\sigma=0.02$	6	1.669	0.225	0.3333	1.876
noise	reset $\sigma=0.05$	6	1.649	0.25	0.3333	1.867
noise	reset $\sigma=0.1$	6	1.679	0.225	0.3333	1.847
noise	reset $\sigma=0.2$	6	1.702	0.2	0.3333	1.849
sparsity	0.0	6	0.8142	0.375	0.1667	2.175
sparsity	0.001	6	0.8433	0.475	0.1667	2.21
sparsity	0.01	6	0.8875	0.525	0.1667	2.085
sparsity	0.05	6	0.2217	0.8	0.1667	2.051
sparsity	0.2	6	0.4741	0.65	0.1667	2.033
sparsity	1.0	6	1.01	0.6	0.1667	2.003
optimizer	bfgs_propagate	8	0.2053	0.95	0.125	1.439
optimizer	bfgs_reset	8	1.648	0.225	0.625	13.55
optimizer	lbfgs_propagate	8	0.2053	0.95	0.125	102.4
optimizer	lbfgs_reset	8	1.648	0.225	0.625	15.25
iters	250	4	1.722	0.325	0.25	0.431
iters	500	4	1.426	0.2	0.25	0.8172
iters	1000	4	0.6087	0.575	0.25	1.047
iters	2500	4	0.2658	0.775	0.25	2.014
iters	5000	4	0.1986	0.9	0.25	3.867
substeps	1	4	0.4988	0.75	0	0.2064
substeps	2	4	0.3377	0.675	0.25	0.4289
substeps	5	4	0.2853	0.825	0.25	1.027
substeps	10	4	0.2658	0.775	0.25	1.991
substeps	20	4	0.2658	0.775	0.25	4.059
seedscale	sc0.01_propagate_flat	8	0.1967	0.85	0	2.064
seedscale	sc0.01_propagate_graded	8	0.1967	0.85	0	2.062
seedscale	sc0.01_reset_flat	8	1.65	0.35	0.125	1.951
seedscale	sc0.01_reset_graded	8	1.589	0.375	0	1.88
seedscale	sc0.03_propagate_flat	8	0.2857	0.775	0	2.085
seedscale	sc0.03_propagate_graded	8	0.2857	0.775	0	2.084
seedscale	sc0.03_reset_flat	8	1.586	0.475	0.25	1.875
seedscale	sc0.03_reset_graded	8	1.573	0.35	0	1.934
seedscale	sc0.1_propagate_flat	8	0.2086	0.85	0.125	2.025
seedscale	sc0.1_propagate_graded	8	0.2086	0.85	0	2.047
seedscale	sc0.1_reset_flat	8	1.649	0.25	0.25	1.888
seedscale	sc0.1_reset_graded	8	1.631	0.3	0	1.864
seedscale	sc0.3_propagate_flat	8	0.4237	0.7	0.25	2.043
seedscale	sc0.3_propagate_graded	8	0.3495	0.825	0.125	2.179
seedscale	sc0.3_reset_flat	8	2.443	0.1	0.25	1.875
seedscale	sc0.3_reset_graded	8	2.702	0.1	0.25	1.831

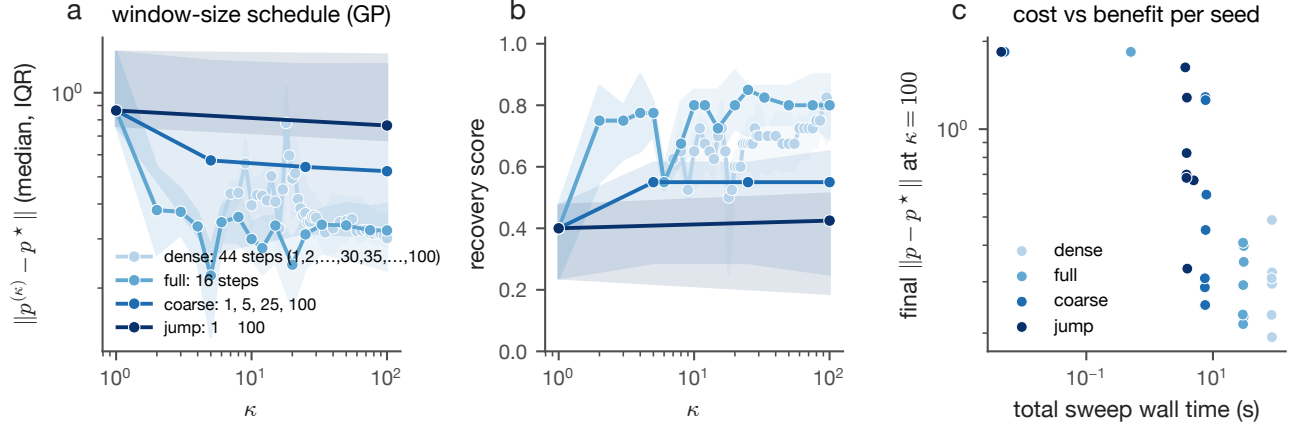


Figure 21: Window-size schedule (how many nodes to remove per step). GP with the DENSE (44 stages), FULL (16), COARSE (4) and JUMP (2: $\kappa = 1$ then 100) schedules, 8 seeds each: (a) $\|p - p^*\|$ and (b) score along the sweep (median, IQR band), (c) final error against total wall time per seed. Final medians: dense 0.30, full 0.32, coarse 0.52, jump 0.76 (scores 0.80, 0.80, 0.55, 0.43). Removing few nodes per step is what makes propagation work — the manuscript’s design rule — and the dense schedule costs $2.7\times$ the time of the full one for no gain. Data: `sweeps.csv` (`exp=schedule`, `main/propagate`).

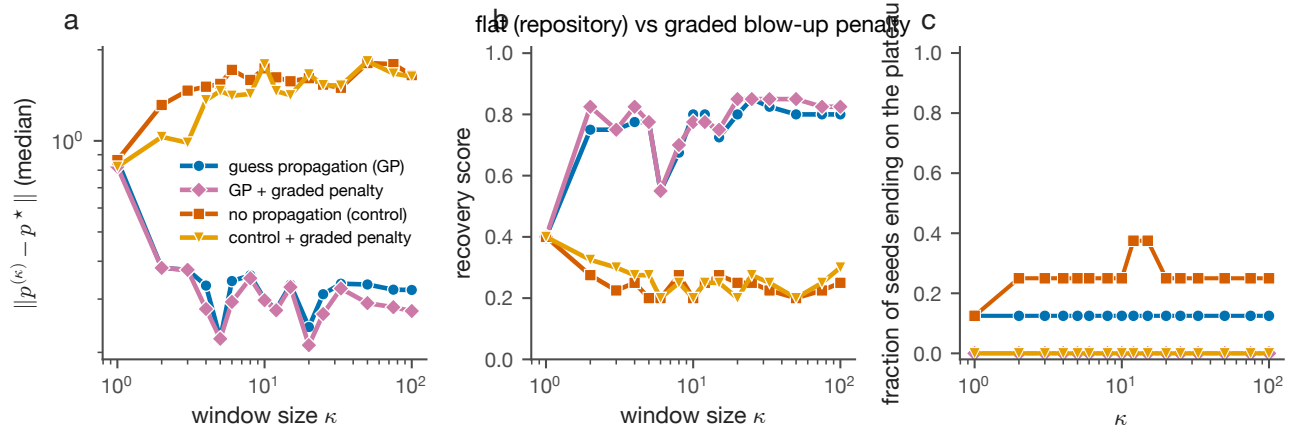


Figure 22: Flat versus graded blow-up penalty. The repository returns the constant 10^3 on blow-up; the graded variant adds a term that decreases with the index at which the blow-up occurred. GP with the graded penalty (purple) ends at $\|p - p^*\| = 0.27$, score 0.83, with *no* seed on the plateau at any stage (c), versus 0.32 / 0.80 / 12.5% for the flat penalty (blue): seed 1 is rescued. For the control the penalty changes nothing (still ≈ 1.6): without propagation the failure is the landscape, not the plateau. Data: `sweeps.csv` (`exp=penalty`, `main`).

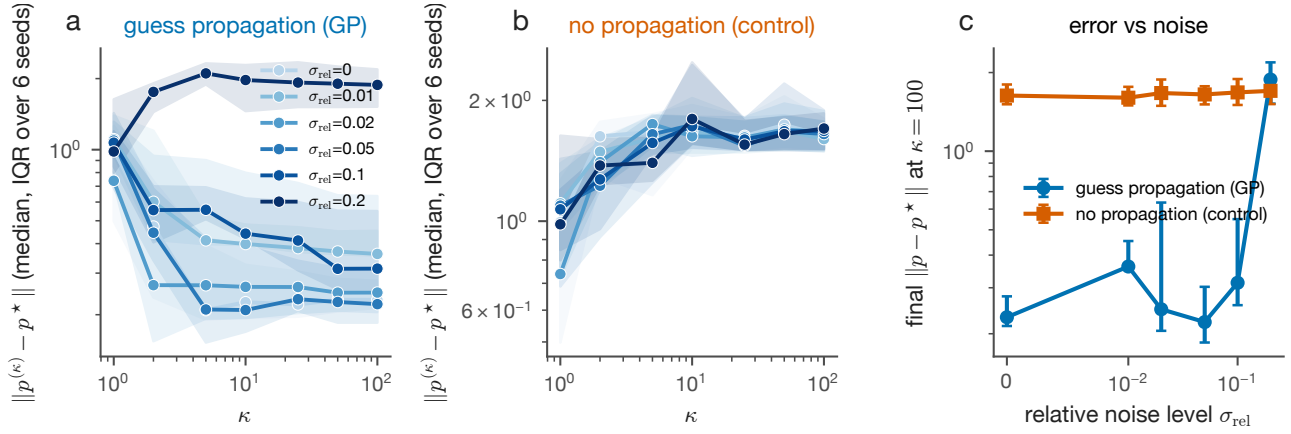


Figure 23: Observation noise. GP (a) and control (b) along the SHORT schedule for relative noise $\sigma_{\text{rel}} \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2\}$ (6 seeds each; the data-noise seed is fixed so the noise realisations are scaled versions of one another); (c) final error at $\kappa = 100$. GP ends at 0.22–0.36 for $\sigma_{\text{rel}} \leq 0.1$ and fails at 0.2 (1.88); the control ends at 1.6–1.7 at *every* noise level including 0 — restarting from the seed fails because of the landscape, not the noise. Note that the GP error does not go to zero at zero noise (0.23): the remaining error is the sparsity-term shrinkage and Nelder–Mead’s finite accuracy, not noise. Data: `sweeps.csv` (`exp=noise`).

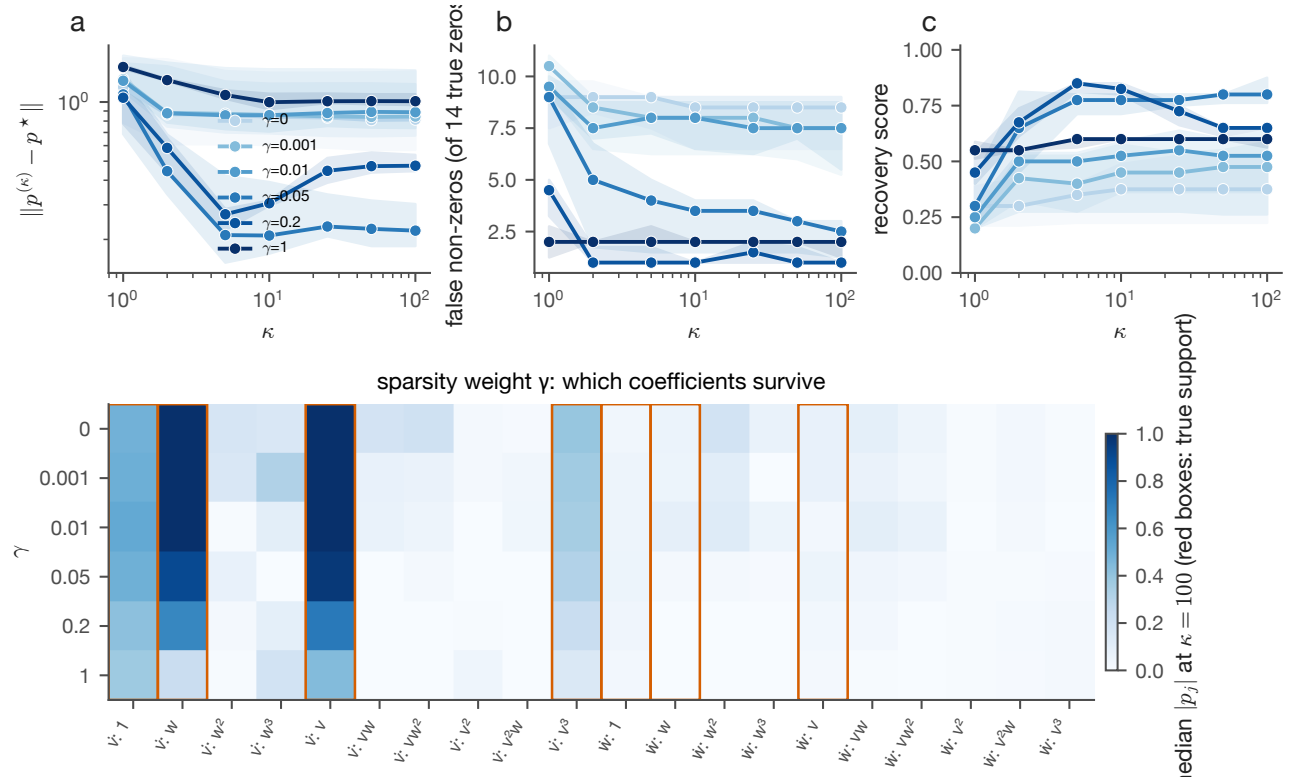


Figure 24: Sparsity weight γ . GP along the SHORT schedule for $\gamma \in \{0, 10^{-3}, 10^{-2}, 0.05, 0.2, 1\}$, 6 seeds: (a) $\|p - p^*\|$, (b) false non-zeros, (c) score; bottom: median $|p_j|$ at $\kappa = 100$ per coefficient (red boxes = true support). $\gamma = 0.05$ (the repository default) is the best of the six (0.22, score 0.80); with $\gamma \leq 0.01$ the Nelder–Mead sweep wanders into dense, wrong models (0.81–0.89, 4–6 false non-zeros), and $\gamma \geq 0.2$ over-shrinks the true coefficients (0.47, 1.01). Data: `sweeps.csv`, `sweep_minimizers.csv` (`exp=sparsity`).

optimiser: derivative-free vs AD-gradient (solid: GP, dashed: no propagation)

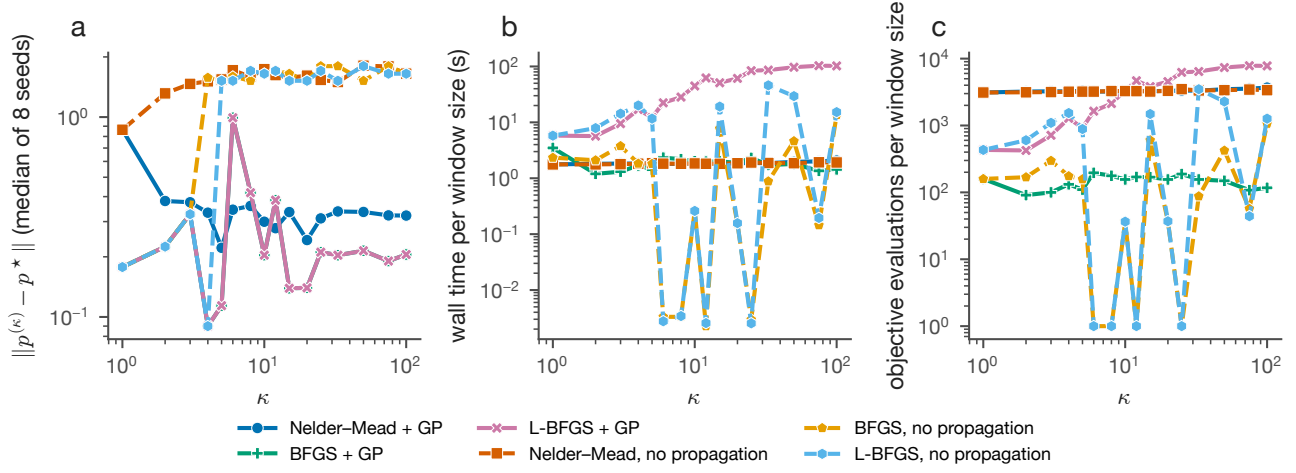


Figure 25: Optimiser: derivative-free versus AD gradient. Nelder-Mead, BFGS and L-BFGS (exact ForwardDiff gradient of the windowed loss, possible after fix F4), with (solid) and without (dashed) propagation, 8 seeds, FULL schedule: (a) median $\|p - p^*\|$, (b) wall time per stage, (c) objective evaluations per stage. BFGS+GP is the best arm (0.21, score 0.95) and needs ≈ 150 evaluations per stage instead of 3500; L-BFGS reaches the same minimiser but hits the iteration cap at large κ (100s per stage). Without propagation the gradient methods stop after one evaluation whenever the start is on the plateau (zero gradient) — 62–71 % of their stages — and otherwise converge to spurious minima. Data: `sweeps.csv` (`exp=optimizer, main`).

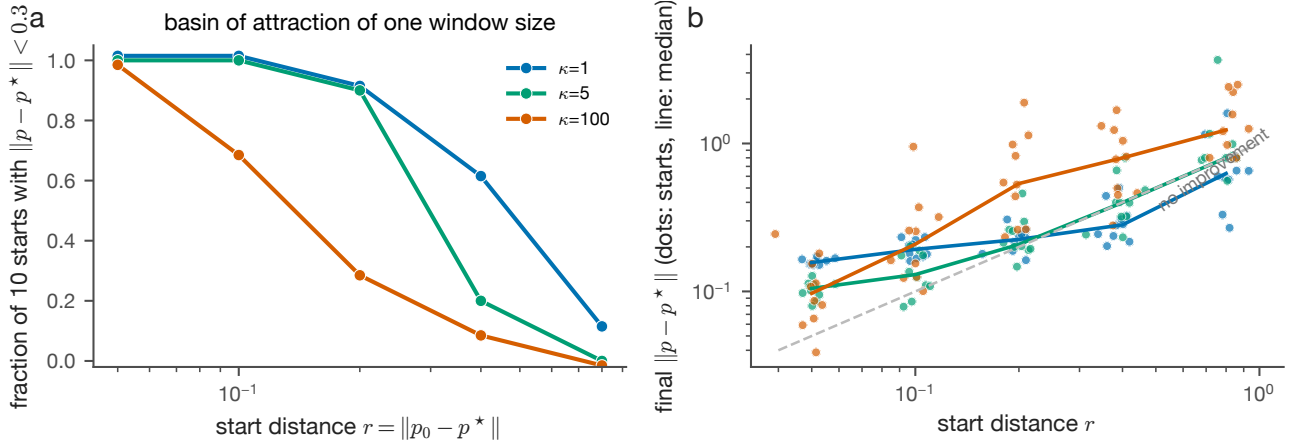


Figure 26: Basin of attraction of a single window size. One Nelder-Mead run (2500 iterations) of J_κ from $p_0 = p^* + r u$ (10 random unit directions per radius $r \in \{0.05, \dots, 0.8\}$) for $\kappa = 1, 5, 100$: (a) fraction of starts ending within 0.3 of p^* , (b) final error (dots) and median. Single shooting ($\kappa = 100$) only converges from $r \leq 0.1$; $\kappa = 5$ from $r \leq 0.2$; $\kappa = 1$ never gets below ≈ 0.15 (its noise/shrinkage floor) but improves every start up to $r = 0.4$. This is the quantitative version of 6: small windows have the wide basin, large windows have the accurate minimum. Data: `sweeps.csv` (`exp=basin`).

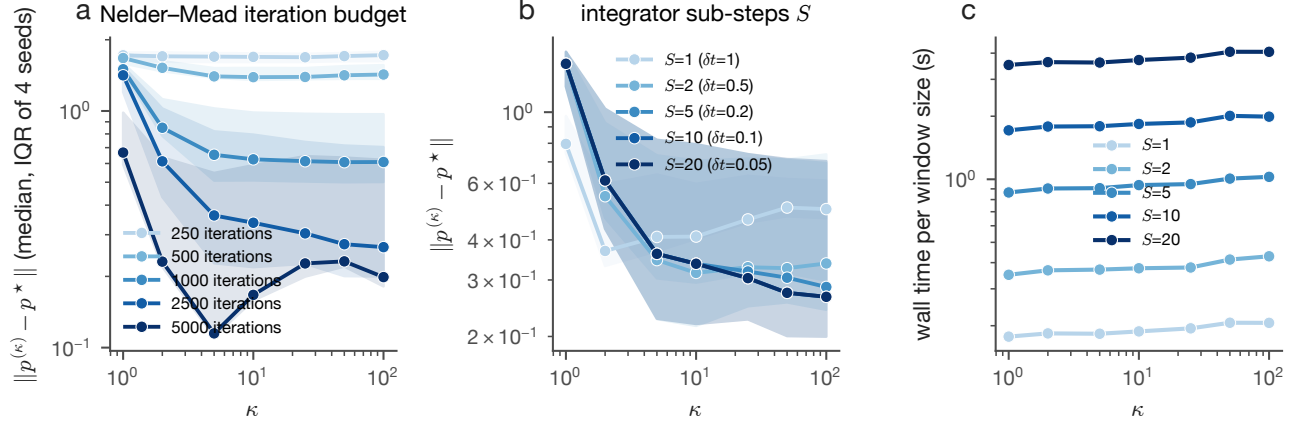


Figure 27: Iteration budget and integrator resolution. (a) GP with 250–5000 Nelder–Mead iterations per stage (4 seeds, SHORT): final medians 1.72, 1.43, 0.61, 0.27, 0.20 — fewer than 2500 iterations in 22 dimensions is not converged; (b) sub-steps $S \in \{1, 2, 5, 10, 20\}$ per data interval ($\delta t = 1/S$): $S \geq 5$ gives the same answer (0.27–0.29), $S = 1$ ($\delta t = \Delta t$) biases the fit (0.50) but never blows up; (c) wall time scales linearly with S . Data: `sweeps.csv` (`exp=iters`, `exp=substeps`).

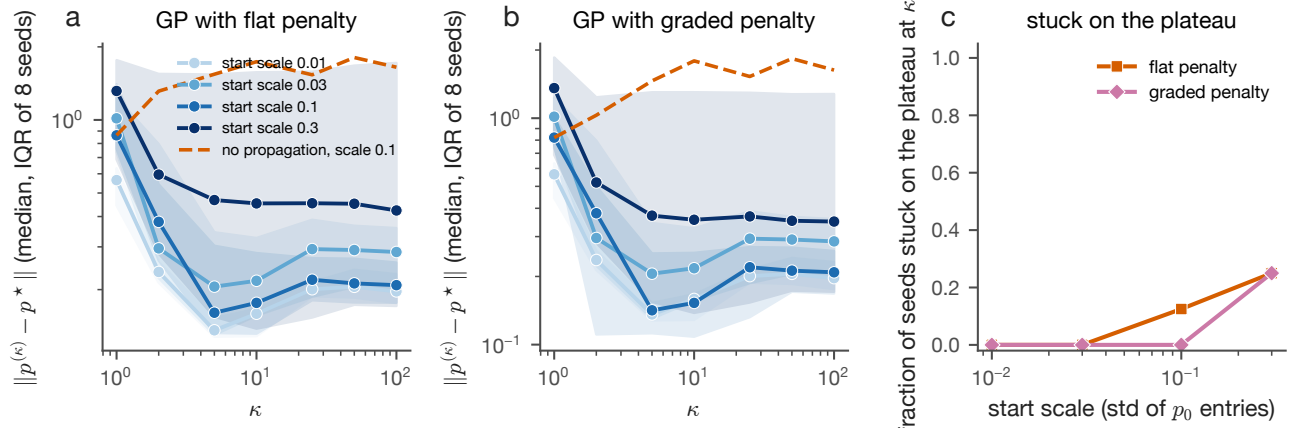


Figure 28: Start distance. GP along the SHORT schedule from $p_0 = \text{scale} \times \xi$ with scale 0.01 (the repository’s driver), 0.03, 0.1 (R001 and this report) and 0.3, 8 seeds, with the flat (a) and graded (b) penalty; the dashed reference is the control at scale 0.1; (c) fraction of starts stuck on the plateau at $\kappa = 1$. Final medians: 0.20, 0.29, 0.21, 0.42 (flat) and 0.20, 0.29, 0.21, 0.35 (graded). With the flat penalty 12.5–25% of the starts at scale ≥ 0.1 never leave the plateau; the graded penalty reduces that to 0–12.5%. Data: `sweeps.csv` (`exp=seedscale`).

9 Part F — Integrators and solvers inside the loss

Table 4: Loss evaluation at p^* , $\kappa = 5$, $\gamma = 0$, for 14 integrator settings inside the windowed loss (in-house fixed-step Tsit5 with S sub-steps; DifferentialEquations.jl solvers through the frozen mod 8 stiff branch); reference = adaptive Tsit5 at tolerance 10^{-12} . Time = median of 5 evaluations. From `solvers_eval.csv`.

method	setting	J	$ J - J_{\text{ref}} $	ms per eval
reference_diff_eq_Tsit5_adaptive	abstol=reltol=1e-12	0.0132220	0	2.296
inhouse_Tsit5_fixed	S=1	0.2818782	0.2687	0.05596
inhouse_Tsit5_fixed	S=2	0.0132233	1.262e-06	0.1089
inhouse_Tsit5_fixed	S=5	0.0132220	1.681e-10	0.2711
inhouse_Tsit5_fixed	S=10	0.0132220	1.359e-13	0.5429
inhouse_Tsit5_fixed	S=20	0.0132220	9.17e-15	1.085
inhouse_Tsit5_fixed	S=50	0.0132220	2.382e-15	2.826
diff_eq_Tsit5_adaptive	abstol=reltol=1e-4	0.0132225	5.223e-07	0.2142
diff_eq_Tsit5_adaptive	abstol=reltol=1e-8	0.0132220	2.201e-10	0.5102
diff_eq_Rosenbrock23_adaptive	abstol=reltol=1e-8	0.0132218	1.747e-07	5.689
diff_eq_Rodas5_adaptive	abstol=reltol=1e-8	0.0132220	3.545e-10	1.976
diff_eq_RadauIIA5_adaptive	abstol=reltol=1e-8	0.0132220	7.083e-09	1.327
diff_eq_ImplicitEuler_fixed	dt=0.1	0.0120178	0.001204	1.537
diff_eq_Tsit5_fixed	dt=0.1	0.0132220	1.364e-13	0.7591

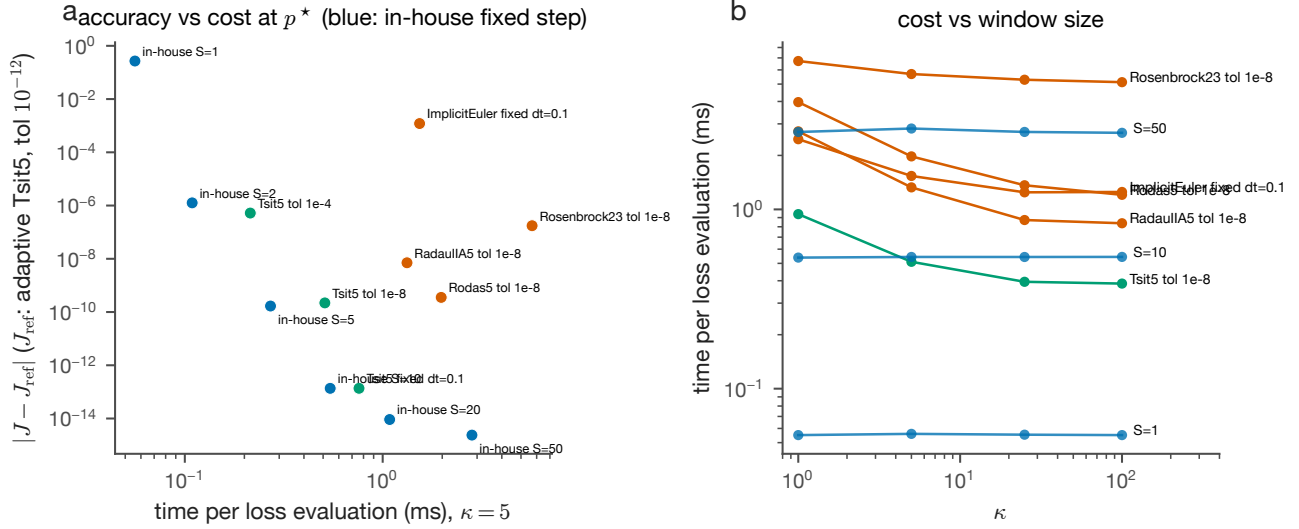


Figure 29: Accuracy versus cost of the integrator inside the loss. (a) $|J - J_{\text{ref}}|$ against the time per evaluation at $\kappa = 5$, p^* , for the 14 settings of Table 4. The in-house fixed-step Tsit5 at $S = 10$ (the default) agrees with the reference to 1.4×10^{-13} at 0.54 ms; the adaptive DifferentialEquations Tsit5 at 10^{-8} is as accurate at the same cost; implicit solvers (Rosenbrock23, Rodas5, RadauIIA5) cost 3–10 $\times$ more for no gain — the problem is not stiff; only $S = 1$ (2.7×10^{-1}) and fixed-step implicit Euler (1.2×10^{-3} , 9% of J) distort the objective. (b) Time per evaluation versus κ for a subset. Data: `solvers_eval.csv`.

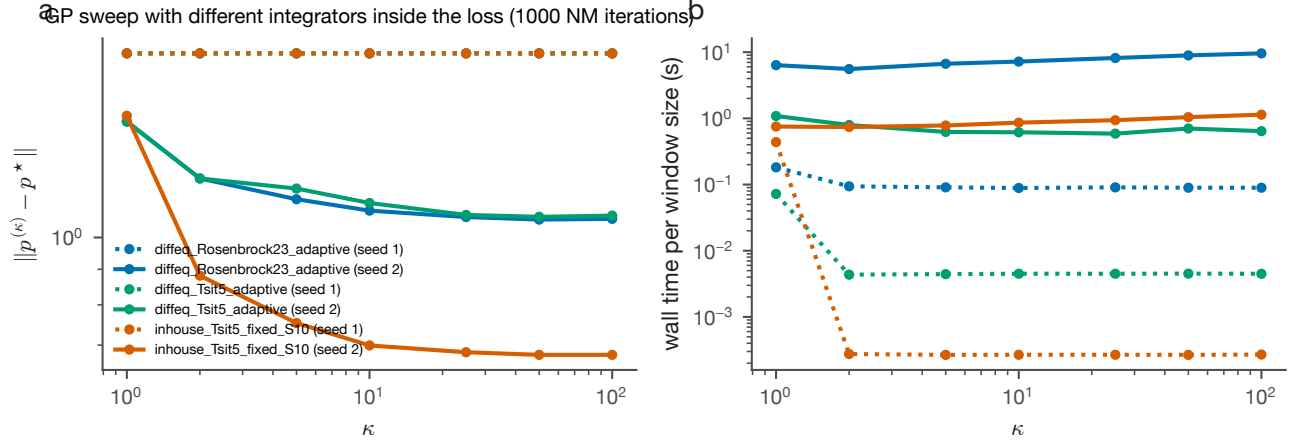


Figure 30: GP sweep with three integrators inside the loss. In-house Tsit5 ($S = 10$), adaptive DifferentialEquations Tsit5 (10^{-8}) and Rosenbrock23 (10^{-8}), SHORT schedule, 1000 Nelder–Mead iterations, seeds 1 (dotted) and 2 (solid): (a) $\|p - p^*\|$, (b) wall time per stage. Seed 1 is the plateau start and fails identically in all three (1.84 at every stage, 0 iterations); seed 2 converges with all three, the two DifferentialEquations arms agreeing exactly at $\kappa \leq 2$ and diverging afterwards through the chaotic sensitivity of a derivative-free optimiser to 10^{-10} differences in the objective (final 0.68 vs 1.06–1.08 at only 1000 iterations). Rosenbrock23 costs $8\times$ the time. Data: `solvers_sweep.csv`.

10 Part G — Defects found and fixed

Each defect is quantified by executing the frozen pre-fix source (`deps/prefix/`) against the frozen post-fix source (`deps/`), never by re-typing; `analysis/06_bugs.jl`. Table 5 summarises. The fixes are in the live tree in the same commit as this report (every changed line carries the comment R002 fix):

- **F1** (R001; mod 6, `odefun_new`): the continuation lines beginning with `+` were separate, discarded statements, so the Jacobian behind the stability/eigenvalue movies was that of the *linear truncation* of the vector field. Fixed by parenthesising the RHS. All movies under `figs/window_size_plots/stability/` are wrong and should be regenerated.
- **F2** (R001; mod 7 stiff branch of `forward_simulation_loss_windows`): the same parsing trap dropped the entire w -residual. Fixed by parenthesising. (Mod 8 already used a matrix norm and was not affected.)
- **F4** (R001; mod 7 and mod 8 drivers): `grad_fs_loss_window!` differentiated the single-shooting loss `fs_loss`, not `fs_loss_window`. Fixed; this is what makes the BFGS arms of 25 possible.
- **F5** (R001; `plotting_window_size.jl`, `animate_contourplots!`): stale call signature (missing `unstable_param` slot, keyword `silent` instead of `silent_`). Fixed to the mod 6+ signature; *not executed here* (Makie cannot load in the sandbox), so this fix is a reading, not a measurement.
- **F6** (R001; mod 8 driver): the `@threads` loop wrote the shared `x0`, `results_inner` and a Dict. Fixed with per-run copies and a lock; not a figure.
- **N1 (new)** mod 8 driver: the sweep called the pre-refactor `odefun` (order 1, $v, w, \dots$) while the data, `fhn_p` and the polish used the library order; the bar-chart labels `trmstr` encoded the old order and `plotSolution!` was called with an undefined name `trm`. The truth vector read in the wrong order is 2.50 away from itself (34a) and its cost at $\kappa = 1$ is 0.68 instead of 0.018: the committed sweep could not have recovered the model. Fixed (sweep and plots use `odefun_generic`; labels generated from `multiindex_mapping`).
- **N2 (new)** mod 8, `forward_simulation_loss` (explicit branch, used by the final BFGS polish): read the *global data* instead of its `data_` argument, so the function ignored the data it was given (34b). Fixed. This is the shadowing R001 asked to check when the refactor landed.
- **N3 (new, behavioural)** the committed driver had guess propagation *disabled* (`copyto!(x0, x0_)` after every stage). Replaced by an explicit `guess_mode` switch (`:propagate` default, `:best`, `:reset`).

- Minor: undefined `y0` default in `custom_integrator`; wrong bound in the window-size warning.

R001’s bug-presence gates G3, G4, G5 and G7 are retired by this report: R002’s gates G5–G9 assert the defects are present in `deps/prefix/` and absent in `deps/`.

Table 5: Defect quantification (pre-fix vs post-fix frozen sources). From `bug_summary.json` and the `bug_*.csv` files.

finding	measure	value
F2 (R001): mod 7 stiff branch drops the w -residual	share of $J(p^*)$ missing, $\kappa=1/10/25/100$	11.5 % / 9.3 % / 17.0 % / 11.2 % / 11.9 %
F1 (R001): mod 6 <code>odefun_new</code> = linear truncation	$\partial\dot{v}/\partial v$ at $u=(1.5,0.4)$: pre-fix vs fixed	1 vs -1.25
F4 (R001): sweep gradient = single-shooting gradient	$\cos\angle(g\kappa, g100)$ at $\kappa=1 / 5$	0.180 / 0.989
NEW: mod 8 driver mixed monomial orders	spurious $\ p - p^*\ $ between orderings	2.501
NEW: mod 8 forward_simulation_loss ignores <code>data_</code>	J identical for 3 datasets pre-fix / differs post-fix	True / True

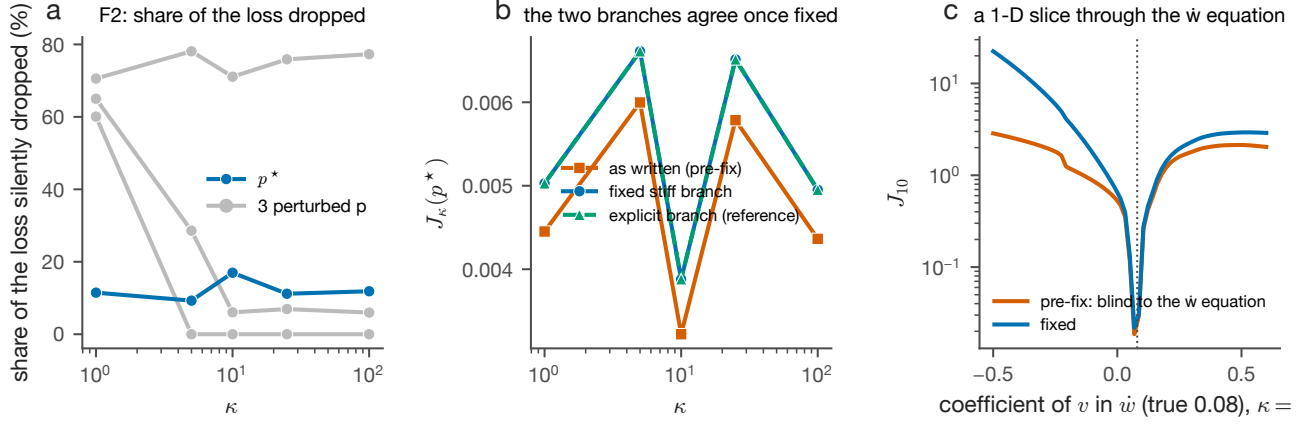


Figure 31: F2: the mod 7 stiff branch silently discarded the w -residual. (a) Share of the loss missing, $(J_{\text{fixed}} - J_{\text{prefix}})/J_{\text{fixed}}$, at p^* (blue: 11.5, 9.3, 17.0, 11.2, 11.9% at $\kappa = 1, 5, 10, 25, 100$, $\gamma = 0$) and for three perturbed parameter vectors (grey; up to 78% — away from the optimum the w trajectory is badly wrong and the dropped term dominates; rows at 0 are blow-ups, where both versions return 10^3). (b) The three implementations at p^* : pre-fix stiff branch, fixed stiff branch and the explicit in-house branch agree to 3×10^{-10} once fixed. (c) A 1-D slice through the v coefficient of the $\dot{w}$ equation at $\kappa = 10$: the fixed objective varies by a factor 10^3 across the slice, the pre-fix one barely moves — the stiff branch was blind to the $\dot{w}$ equation, so every contour/surface movie made with `stiff_solver=true` shows the wrong landscape. Data: `bug_F2_w_residual.csv`, `bug_F2_slice.csv`.

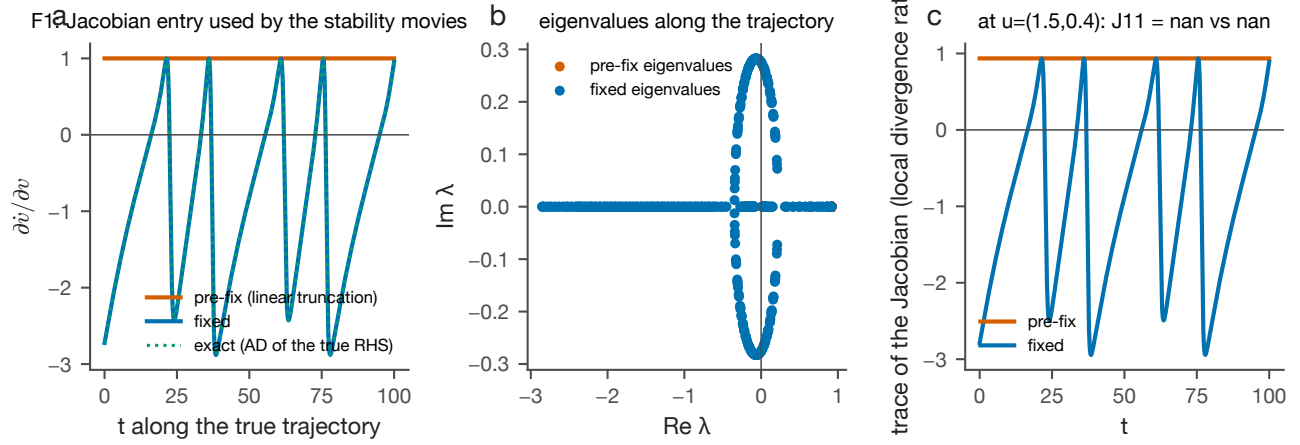


Figure 32: F1: the stability movies used a linear truncation of the vector field. Along the true FHN trajectory (401 points): (a) $\partial \dot{v}/\partial v$ from the pre-fix `odefun_new` (vermilion, constant +1), the fixed one (blue) and the exact Jacobian of the true RHS (green dotted, identical to fixed to 0.0); at $u = (1.5, 0.4)$ the entry is +1.0 instead of -1.25 — the wrong sign. (b) Jacobian eigenvalues along the orbit, pre-fix (a fixed pair) versus fixed (moving with the state, crossing into the stable half-plane on the slow branches). (c) The trace (local divergence rate), constant and positive pre-fix, mostly negative after the fix. Data: `bug_F1_jacobian.csv`, `bug_F1_meta.json`.

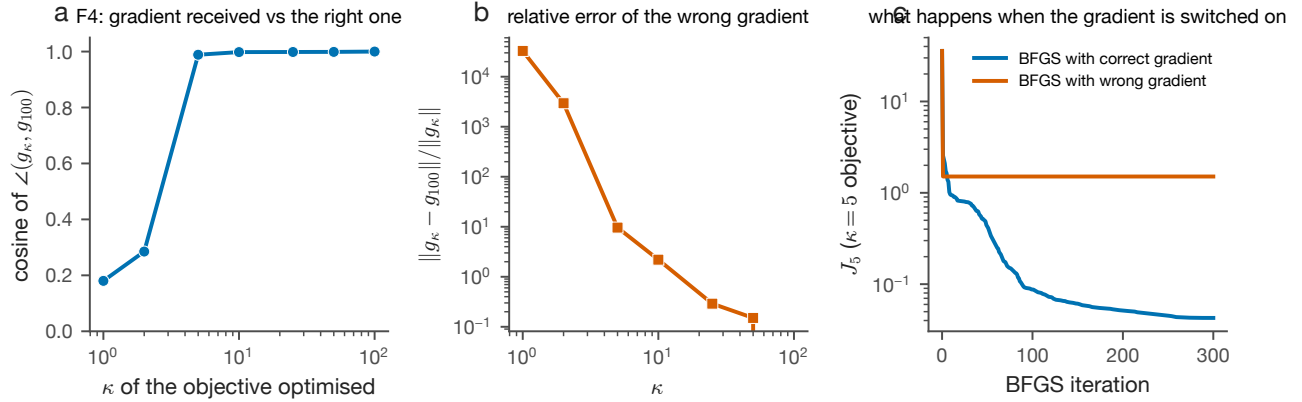


Figure 33: F4: the sweep's gradient was the gradient of the wrong function. At $z = [y_0; p^* + 0.05 \xi]$: (a) cosine between the gradient of the κ -windowed objective and the single-shooting gradient the pre-fix code returned — 0.18 at $\kappa = 1$ (80°), 0.99 at $\kappa = 5$, 1 at $\kappa = 100$; (b) the relative error of the wrong gradient, 3×10^4 at $\kappa = 1$ (the single-shooting gradient is huge because that trajectory has diverged). (c) 300 BFGS iterations on the $\kappa = 5$ objective from the same start with the correct gradient (J_5 : 35.8 $\rightarrow$ 0.043) and with the wrong one (stalls at 1.51 from iteration 19, 23 708 line-search evaluations). The defect was harmless only because the sweep used Nelder–Mead, which ignores the gradient. Note: this start ($\|p - p^*\| = 0.23$) has a large cost because the 0.05 perturbation hits the cubic term; the correct-gradient run lowers J_5 but ends at $\|p - p^*\| = 3.96$ (a different, wrong minimum) — the panel demonstrates the gradient, not recovery. Data: `bug_F4_gradient.csv`, `bug_F4_bfgs.csv`.

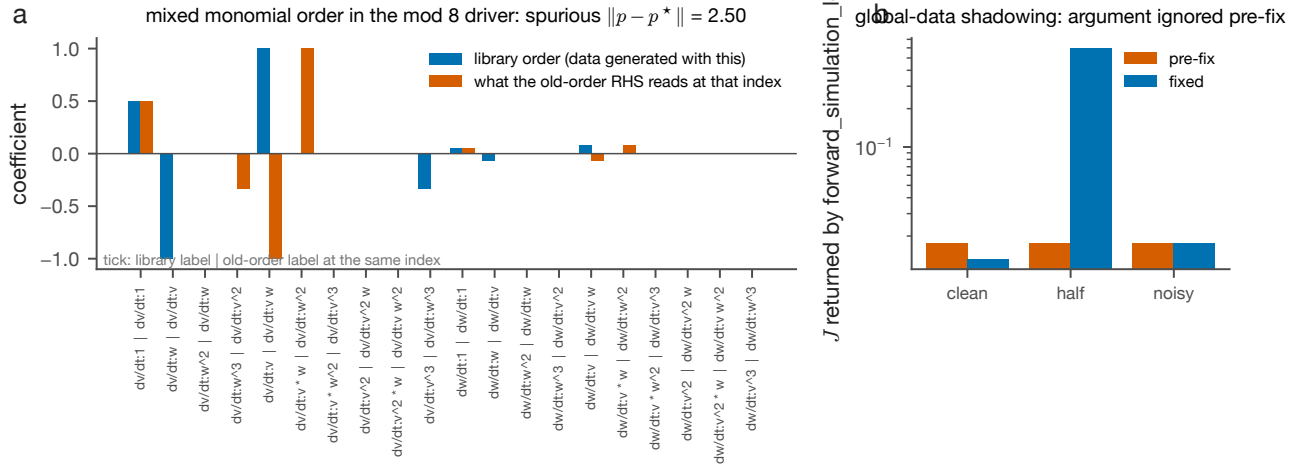


Figure 34: New defects in the committed mod 8 driver. (a) N1, mixed monomial order: the true coefficients in the library order the data were generated with (blue) and what the old-order RHS used by the sweep reads at the same index (vermilion); tick labels give both names. The two readings of the same vector are 2.50 apart (larger than $\|p^*\| = 1.54$), and the true vector evaluated through the old-order RHS costs 0.68 ($\kappa = 1$), 2.06 ($\kappa = 5$), 7.83 ($\kappa = 100$) instead of 0.018–0.021. (b) N2, global-data shadowing: `forward_simulation_loss` called with three different data matrices (noisy, clean, states halved) returns the same 0.0176 pre-fix and 0.0176 / 0.0133 / 0.592 after the fix. Data: `bug_ordering.csv`, `bug_ordering_meta.json`, `bug_shadowing.csv`.

11 Part H — Other systems and closing the loop with the theory

The core is dimension- and degree-agnostic; two further systems test it (Table 6). Lotka–Volterra: $\dot{x} = x - 0.5xy$, $\dot{y} = -0.8y + 0.3xy$, quadratic library (6 monomials, 12 coefficients), $N = 101$, $\Delta t = 0.25$, 5% noise, Nelder–Mead. Lorenz: $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, quadratic library (10 monomials, 30 coefficients), $N = 101$, $\Delta t = 0.02$ ($t \in [2, 4]$ of a trajectory from $(-8, 7, 27)$), 2% noise, start scale 0.5; BFGS (2500-iteration cap) and Nelder–Mead (4000). Same SHORT schedule and $\gamma = 0.05$.

Table 6: Other systems at $\kappa = 100$: seed medians of $\|p - p^*\|$ and score, and the fraction of seeds on the plateau. From `other_sweeps.csv`.

system	optimizer	arm	n_seeds	$\ p - p^*\ $ ($\kappa=100$)	score	plateau frac
lorenz	bfgs	propagate	4	3.852	0.8333	0
lorenz	bfgs	reset	4	31.95	0.03333	1
lorenz	nm	propagate	4	18.11	0.2333	0
lorenz	nm	reset	4	31.95	0.03333	1
lv	nm	propagate	6	0.29	0.8333	0
lv	nm	reset	6	1.408	0.2917	0.8333

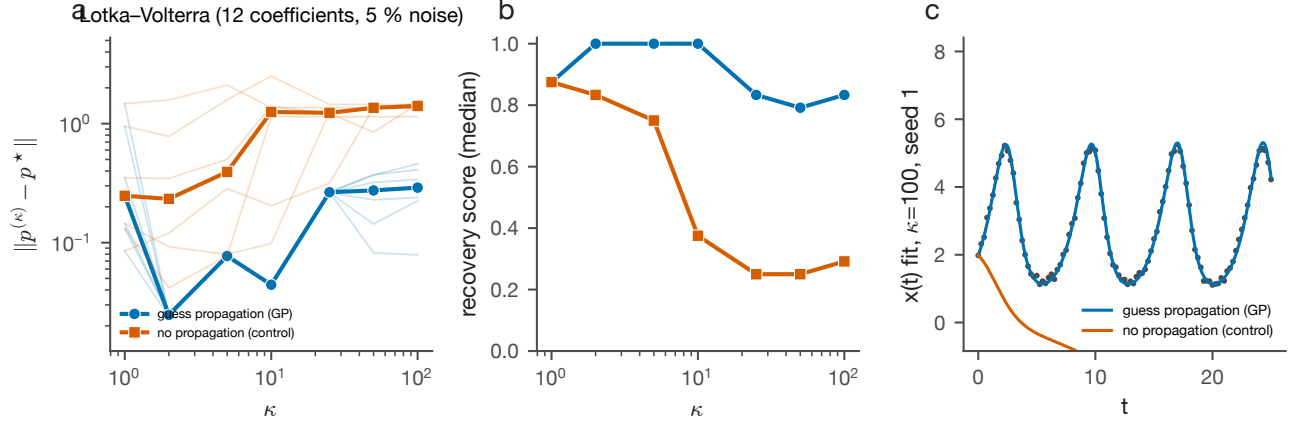


Figure 35: Lotka–Volterra. (a) $\|p - p^*\|$ along the sweep (thin: 6 seeds, thick: median), (b) score, (c) the $x(t)$ fit at $\kappa = 100$ for seed 1. GP ends at 0.29 (score 0.83; one seed reaches 0.025 with score 1.0 at $\kappa = 2$ –10, i.e. exact recovery), the control at 1.41 with 5 of 6 seeds on the plateau; GP is better in every seed. Data: `other_sweeps.csv`, `other_fits.csv` (`system=lv`).

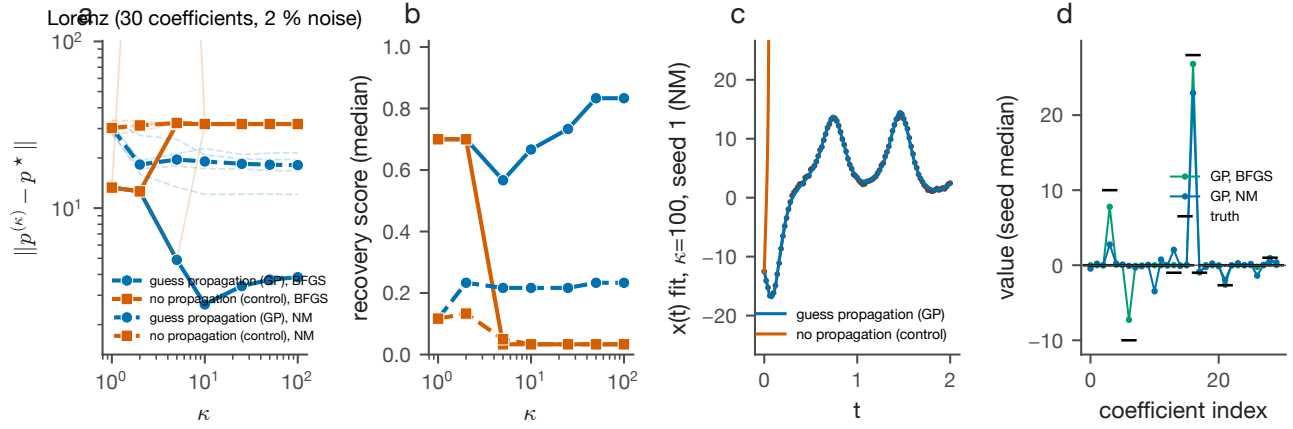


Figure 36: Lorenz (chaotic, 30 coefficients). (a) $\|p - p^*\|$ along the sweep for BFGS (solid) and Nelder–Mead (dashed), GP (blue) and control (vermillion), 4 seeds; (b) score; (c) the $x(t)$ fit at $\kappa = 100$, seed 1; (d) the seed-median GP coefficients versus truth. GP+BFGS recovers Lorenz ($\|p - p^*\| = 3.9$ on coefficients of size up to 28, score 0.83, no blow-up); GP+Nelder–Mead never converges within 4000 iterations in 33 dimensions (18.1, score 0.23) — a derivative-free sweep does not scale; the control blows up in 4 of 4 seeds from $\kappa = 5$ on with either optimiser. Data: `other_sweeps.csv`, `other_minimizers.csv`, `other_fits.csv` (`system=lorenz`).

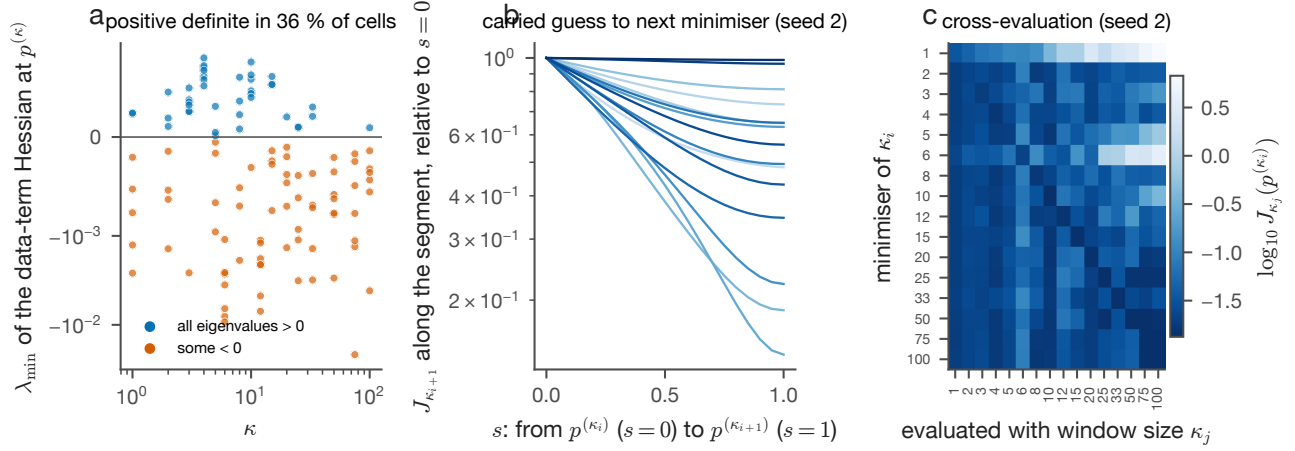


Figure 37: The theory’s premises checked where they are assumed. (a) Smallest eigenvalue of the data-term Hessian at the GP minimiser $p^{(\kappa)}$ returned by Nelder–Mead at every stage, 7 seeds (the plateau seed excluded; symlog scale; blue = all 20 eigenvalues positive, vermilion = at least one negative). Where the manuscript assumes local strong convexity — at the minimiser, not at p^* — the returned point is positive definite in only 36 % of the (seed, κ) cells, with on average 0.8 of 20 eigenvalues slightly negative elsewhere ($|\lambda_{\min}| \lesssim 10^{-3}$ against $\lambda_{\max}$ of 10 – 10^6): Nelder–Mead stops short of a true stationary point in 22 dimensions (cf. 27a), so the premise is *not* verified at the returned iterates; the BFGS iterates of 25 would be the right place to test it. (b) The cost of the *next* stage along the straight segment from $p^{(\kappa_i)}$ to $p^{(\kappa_{i+1})}$ (seed 2), relative to its value at the carried guess: it decreases monotonically along every segment, i.e. the carried guess lies inside the basin of the next minimiser — the mechanism the node-removal theorem describes. (c) Cross-evaluation (seed 2): the minimiser of κ_i evaluated at every other κ_j (log colour); the near-diagonal structure shows that a minimiser is good only for nearby window sizes, which is why the schedule must be gradual (21). Data: `post_hessian_at_minimizers.csv`, `post_path_between_minimizers.csv`, `post_cross_evaluation.csv`.

12 Part I — Extended landscape animations: plane selection, resolution, and 3-D

The landscape animation of Part C (`anim_fhn_landscape.mp4`) shows one FHN coefficient plane at 61×61 . This part extends it to five FHN planes and one Lotka–Volterra plane at 201×201 , adds a rotating 3-D version of four of them, and — because “pick an interesting plane” is exactly the kind of choice that invites cherry-picking — states the rule by which the planes were chosen and shows the evidence for every candidate, selected or not.

12.1 Definitions used in this part

Everything below is defined here; nothing is inherited implicitly.

- **Coefficient plane** $\Pi(i, j)$. The two-dimensional slice of parameter space on which the coefficients p_i and p_j sweep a rectangle centred on their true values p_i^*, p_j^* while *all other coefficients are held at the truth* (18 of them for FHN, 10 for Lotka–Volterra) and the initial state is the first datum. A plane is named by its two library monomials and the state equation each sits in, e.g. $\Pi(v \text{ in } \dot{v}, v^3 \text{ in } \dot{v})$. Parameter indices follow the library ordering of §2: $1, w, w^2, w^3, v, vw, vw^2, v^2, v^2w, v^3$ for the $\dot{v}$ block (p_1 – p_{10}), then the same ten monomials for the $\dot{w}$ block (p_{11} – p_{20}).
- **Cost** J_κ . Exactly the multiple-shooting cost of §2 with $\gamma = 0.05$, $S = 10$, $\delta t = 0.1$ and window size κ ; it returns the flat value 10^3 on finite-time blow-up.
- **Plateau fraction** φ_κ . The fraction of the plane’s grid cells with $J_\kappa \geq 10^3$: the share of the plane on which the model blows up before the record ends.
- **Basin fraction** β_κ . The fraction of grid cells with $J_\kappa < 2 \min_\Pi J_\kappa$ — the visible area of the valley around the best point of the plane. It is an *area*, not a width: β falling by $10\times$ means the valley floor is $10\times$ smaller

in area, roughly $3\times$ narrower in each direction.

- **Dynamic range** $D_\kappa \cdot \log_{10}(\max J_\kappa / \min J_\kappa)$ over the cells that do *not* blow up, in decades. Small D means a nearly flat plane: the cost barely distinguishes the truth from anything else on it.
- **Rank decorrelation** $1 - \rho_{1,100}$. $\rho_{1,100}$ is the Spearman (rank) correlation between $\log J_1$ and $\log J_{100}$ over the cells that are finite at *both* $\kappa = 1$ and $\kappa = 100$. It is 0 when the two landscapes order the plane identically and 2 when they order it exactly oppositely; a value above 1 therefore means the ranking has *inverted* — points that $\kappa = 1$ prefers, $\kappa = 100$ rejects.
- **Censoring floor.** On the 61×61 screening grid the smallest measurable β is one cell, $1/61^2 = 0.027\%$. Eight of the sixteen candidate planes sit at exactly that value at $\kappa = 100$, so the implied basin ratios β_1/β_{100} of those rows in Table 7 are *lower bounds*, not measurements. Table 8 re-measures the five animated planes on the 201×201 production grid, where the floor is $1/201^2 = 0.0025\%$ and only one cell of the whole table — the basin of $\Pi(w \text{ in } \dot{v}, v \text{ in } \dot{w})$ at $\kappa = 100$ — still sits on it.

12.2 How the planes were chosen

All 16 candidate planes of Table 7 were scanned at 61×61 over the full κ ladder ($\kappa \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 25, 33, 50, 75, 100\}$) by `analysis/12_landscape_planes.jl`; the table is the complete screen, with no candidate omitted. Five planes were animated, each the extreme case of one *distinct* mechanism by which the landscape changes with κ :

- $\Pi(v \text{ in } \dot{v}, v^3 \text{ in } \dot{v})$ — the plane already animated in Part C, kept for continuity and re-rendered at 201×201 . It has the largest rank decorrelation of all sixteen ($1 - \rho = 1.22$): the ordering of the plane inverts between $\kappa = 1$ and $\kappa = 100$.
- $\Pi(v \text{ in } \dot{w}, w \text{ in } \dot{w})$ — the slow block. Largest basin collapse of all sixteen: on the 201×201 production grid β falls from 8.94 % of the plane at $\kappa = 1$ to 0.0050 % (two cells) at $\kappa = 100$ — more than $1800\times$, the largest collapse in the set (Table 8); it also has the largest gain in dynamic range ($1.53 \rightarrow 4.74$ decades). **Read this:** at $\kappa = 1$ this plane is nearly flat — one-interval shooting barely constrains the slow subsystem at all — and by $\kappa = 100$ it is a narrow canyon with a blow-up wall along one side. This is the most drastic transformation in the screen.
- $\Pi(v^2 \text{ in } \dot{v}, v^2 \text{ in } \dot{w})$ — two coefficients that are *zero* in the truth. Largest growth of the blow-up plateau in the screen, φ from 0 % to 57 % of the plane, with D from 2.17 to 4.75 decades. This is the plane on which the spurious quadratic terms that sparse selection must reject are penalised, and it shows that long windows do the rejecting by making those terms *blow up* rather than by making them merely expensive.
- $\Pi(w \text{ in } \dot{v}, v \text{ in } \dot{w})$ — the FHN feedback pair, and the control case: it barely blows up at all. On the 201×201 grid φ is exactly zero for every $\kappa \leq 33$, is a single cell (0.0025 %) at $\kappa = 50$, and reaches only 0.16 % and 0.27 % of the plane at $\kappa = 75$ and $\kappa = 100$, against 28–60 % for the other four planes; yet its dynamic range still grows from 1.93 to 4.54 decades, the largest gain among the planes the screen records with no plateau. Sharpening and plateau growth are separate mechanisms, and this plane isolates the first.
- $\Pi(v^3 \text{ in } \dot{v}, v^3 \text{ in } \dot{w})$ — twin cubic terms. Already rugged at $\kappa = 1$ ($\varphi = 32\%$, $D = 4.71$) and the most blown-up plane at $\kappa = 100$ ($\varphi = 60\%$): ruggedness on top of ruggedness, the least favourable case for an optimiser.

Read this: The selection rule is “extreme on a named mechanism”, not “largest composite score”, and several unselected planes beat selected ones on individual columns: $\Pi(1 \text{ in } \dot{v}, 1 \text{ in } \dot{w})$ and $\Pi(w \text{ in } \dot{v}, w^3 \text{ in } \dot{v})$ have larger (censored) basin ratios than $\Pi(w \text{ in } \dot{v}, v \text{ in } \dot{w})$, and $\Pi(v^3 \text{ in } \dot{v}, vw \text{ in } \dot{v})$ has a larger rank decorrelation than three of the five selected. Table 7 is printed in full so that the choice can be second-guessed.

Table 7: The complete landscape-plane screen (FHN, 61×61 , all 16 candidates, none omitted). Columns as defined in §12.1: rank decorrelation $1 - \rho_{1,100}$, basin fraction β_κ and plateau fraction φ_κ at the two ends of the ladder, and the dynamic range D_κ in decades. A basin of 0.027% is exactly one grid cell — the censoring floor of this grid — so the implied shrink factors of those rows are lower bounds; Table 8 re-measures the animated planes without censoring. “animated” marks the five planes carried to 201×201 . Data: `landscape_plane_screen_summary.csv`, script `analysis/12_landscape_planes.jl`.

plane	truth	1-rho	basin k=1 %	basin k=100 %	plateau k=1 %	plateau k=100 %	decades k=1	decades k=100	animated
v in $\dot{v}$, v^3 in $\dot{v}$	(1, -0.3333)	1.22	0.94	0.027	38	39	4.34	3.67	yes
v in $\dot{v}$, v^2 in $\dot{v}$	(1, 0)	0.54	1.77	0.161	0	0	2.35	3.01	
v^3 in $\dot{v}$, v^2 in $\dot{v}$	(-0.3333, 0)	0.59	1.26	0.215	34	34	4.38	4.58	
w in $\dot{v}$, v in $\dot{v}$	(-1, 1)	0.80	1.29	0.107	0	0	2.09	3.02	
1 in $\dot{v}$, w in $\dot{v}$	(0.5, -1)	0.80	1.26	0.027	0	0	2.20	3.13	
v in $\dot{w}$, w in $\dot{w}$	(0.08, -0.064)	0.46	8.87	0.027	0	28	1.53	4.74	yes
v in $\dot{v}$, v in $\dot{w}$	(1, 0.08)	0.84	2.12	0.027	0	0	2.16	4.37	
w in $\dot{v}$, v in $\dot{w}$	(-1, 0.08)	0.78	2.77	0.027	0	0	1.93	4.54	yes
v^3 in $\dot{v}$, v^3 in $\dot{w}$	(-0.3333, 0)	0.87	0.75	0.027	32	60	4.71	4.31	yes
v^3 in $\dot{v}$, vw in $\dot{v}$	(-0.3333, 0)	0.91	1.64	0.081	32	35	4.55	4.59	
1 in $\dot{v}$, 1 in $\dot{w}$	(0.5, 0.056)	0.62	3.71	0.027	0	0	2.15	2.82	
v^2 in $\dot{v}$, v^2 in $\dot{w}$	(0, 0)	0.77	3.12	0.161	0	57	2.17	4.75	yes
vw in $\dot{v}$, v^2w in $\dot{v}$	(0, 0)	0.68	3.57	0.081	0	35	2.29	3.67	
v in $\dot{v}$, w^3 in $\dot{v}$	(1, 0)	0.76	1.88	0.027	0	13	2.04	3.22	
w in $\dot{v}$, w^3 in $\dot{v}$	(-1, 0)	0.63	8.06	0.054	0	15	1.93	3.47	
v^2 in $\dot{v}$, v^2w in $\dot{v}$	(0, 0)	0.73	3.55	0.188	0	29	2.50	3.70	

12.3 Resolution

The production grids are 201×201 at every one of the 16 window sizes: $10.9\times$ the samples of the 61×61 grid behind the Part C animation and $1.6\times$ the 161×161 Lotka–Volterra grids, i.e. 646 416 cost evaluations per plane. Frames are rendered at 2560×1440 (1440p) rather than the previous 825×616 — $7.2\times$ the pixels. The grids are stored gzipped (`.csv.gz`, read directly by pandas): a plain-text $201 \times 201 \times 16$ grid is about 18 MB and compresses to 3–4 MB. Table 8 reports the geometry of the five animated planes measured on *these* grids; these, not the screening numbers, are what the videos show.

Table 8: The animated planes at production resolution (201×201 ; β_κ and φ_κ as in §12.1). One value is censored, and only one: the basin of $\Pi(w$ in $\dot{v}$, v in $\dot{w}$) at $\kappa = 100$ is a single cell, $1/201^2 = 0.0025\%$, so the shrink factor printed for that row is a lower bound; every other basin in the table is two cells or more and is measured. Column “3-D” marks the four planes that also have a rotating 3-D animation; “key” is the file-name stem — the 2-D video of a row is `anim_fhn_landscape_<key>_hires.mp4` (`anim_lv_landscape_x2_xy_hires.mp4` for the Lotka–Volterra row) and its 3-D video, where it exists, is `anim3d_fhn_landscape_<key>.mp4` (`anim3d_lv_landscape_x2_xy.mp4`). $J_\kappa^{\min}$ is the smallest cost anywhere on the plane at that κ ; it is not in general attained at $p^\star$, because the data carry 5% noise. Data: `landscape_hires_geometry.csv`, written by `figures/make_landscape_animations.py` from the same grids the frames are drawn from.

system	plane	truth	basin k=1 %	basin k=100 %	basin shrink	plateau k=1 %	plateau k=100 %	J min k=1	J min k=100	3-D	key
LV	x^2 in $\dot{x}$, xy in $\dot{x}$	(0, -0.5)	1.337	0.0074	180	0.1	40.2	0.02821	0.02198	yes	lv_x2_xy
FHN	v^2 in $\dot{v}$, v^2 in $\dot{w}$	(0, 0)	3.158	0.0520	61	0.0	56.2	0.01773	0.01737	yes	v2_w2
FHN	v^3 in $\dot{v}$, v^3 in $\dot{w}$	(-0.3333, 0)	0.770	0.0074	104	31.9	59.9	0.01773	0.01398		v3_w3
FHN	v in $\dot{v}$, v^3 in $\dot{v}$	(1, -0.3333)	0.903	0.0124	73	37.4	38.5	0.01755	0.01719	yes	v_v3
FHN	w in $\dot{v}$, v in $\dot{w}$	(-1, 0.08)	2.797	0.0025	1130	0.0	0.3	0.01773	0.01758		w_wv
FHN	v in $\dot{w}$, w in $\dot{w}$	(0.08, -0.064)	8.943	0.0050	1806	0.0	28.2	0.01772	0.01758	yes	wv_ww

12.4 The 3-D animations

Four planes are additionally rendered as a rotating 3-D surface $z = \log_{10} J_\kappa$: the two most drastic FHN planes ($\Pi(v$ in $\dot{w}$, w in $\dot{w}$) and $\Pi(v^2$ in $\dot{v}$, v^2 in $\dot{w}$)), the Part C plane $\Pi(v$ in $\dot{v}$, v^3 in $\dot{v}$) for continuity, and the Lotka–Volterra (x^2, xy) plane of $\dot{x}$. Each animation holds one κ for 16 frames (the first and last for 32) while the camera azimuth advances, completing one full turn over the 288 frames; the guess-propagation and control paths walk over the surface with their shadow on the floor, and the floor carries a filled contour projection of the same surface. **Read this: Display convention.** The 3-D surface is $\log_{10} J_\kappa$ *clipped* at the 99.5th percentile of the finite costs on that plane, and cells on the blow-up plateau are drawn as a flat grey mesa at that ceiling. Without the clip the blow-up value 10^3 — three decades above the bulk of the plane — owns most of the

vertical axis and flattens the valley the picture is about. The clip changes the *height and colour of this one drawn surface* only: no cost value is altered anywhere else in this report, the 2-D animations are unclipped, and the plateau is still identified by colour and named in every frame’s subtitle.

12.5 The same plane without guess propagation

`anim3d_fhn_landscape_wv_ww_control.mp4` is the slow-block 3-D animation with the guess-propagation path deleted: the only trajectory left is the *control* arm, i.e. the minimiser $p^{(\kappa)}$ the optimiser returns at each window size when every stage restarts from the same seed vector rather than from the previous stage’s answer. It is the picture of what the window-size ladder does on its own, before guess propagation is introduced, and it is the natural companion to the two-arm version: the surface tightens into a canyon exactly as before, but the returned minimiser never settles into it — the projected iterate jumps across the plane from one κ to the next, and the gutter panel shows $\|p^{(\kappa)} - p^*\|$ oscillating between 1.43 and 1.95 with no trend along the ladder, on both sides of the shared starting guess (1.52; below it in 6 of the 16 stages) and ending at 1.55 — no better than the 1.44 it already had at $\kappa = 1$. Lengthening the windows alone buys nothing: it is the *carried guess* that converts the tightening landscape into a better parameter estimate. **Read this:** The control minimiser and the minimum of the drawn plane are different objects and must not be read as the same marker. $\arg \min_{\Pi} J_{\kappa}$ is the best point of a slice whose other 18 coefficients are pinned at the truth; on this plane it sits on the truth star at $(0.080, -0.060)$ for every κ in the ladder, to within one grid cell. The control minimiser is a full 20-dimensional optimiser output projected onto the plane, and over the ladder it sits between 0.08 and 0.47 away from the truth. A renderer mode for the former exists (`make_landscape_animations.py 3d-min`) but is not shipped as a video, because a marker pinned to the truth star carries no motion worth animating.

12.6 What the animations do and do not show

- **The surface and the path live in different spaces.** The landscape holds the other 18 (FHN) or 10 (LV) coefficients at the truth, whereas each plotted path point is the projection of a full 20- or 12-dimensional minimiser whose other coefficients are *not* at the truth. A path point can therefore sit visibly off the valley floor of the drawn surface without either being wrong. This caveat applies equally to the Part C animation.
- **Ladder mismatch.** The FHN paths (seed 2) visit all 16 window sizes of the FULL schedule, so the path advances in every frame. The Lotka–Volterra sweep was run on the SHORT schedule, so its path advances only at the window sizes SHORT contains and is stationary in the intervening frames; the surface still morphs at every κ .
- **At $\kappa = 1$ the two arms coincide** by construction — guess propagation has nothing to propagate yet — so the blue marker is hidden under the vermilion one in the first frame.
- **These are slices, not the optimisation landscape.** Nothing here shows the 22-dimensional geometry the optimiser actually traverses; β_{κ} and φ_{κ} are properties of a two-dimensional cut, and the basin collapse they report is not a statement about the volume of the full basin.

13 Gates

`analysis/verify_R002.py` reads only this folder’s own result files and writes `analysis/results/gates_summary.json`. **Status at the final build: 20 of 20 gates pass .** Gates G5–G9 are bug-presence/absence gates (pre-fix present, post-fix absent); they are expected to keep passing as long as `deps/prefix/` is the `44abf4a` snapshot. All other gates are ordinary correctness or completeness gates.

gate	asserts
G1	all 12 frozen files hash to <code>deps/MANIFEST.md</code>
G2	<code>ms_loss</code> = frozen mod 8 loss, relative difference $< 10^{-10}$ (42 evaluations)
G3	dataset = frozen mod 8 data block, $< 10^{-12}$
G4	<code>make_rhs</code> = <code>odefun_poly!</code> and = old-order <code>odefun</code> under the permutation, $< 10^{-12}$
G5	F1 present pre-fix ($\partial\dot{v}/\partial v = +1.0$), absent post-fix (-1.25 = exact, max deviation 0)
G6	F2 present pre-fix (5–25 % of $J(p^*)$ missing at every κ), fixed stiff = explicit to 10^{-8}
G7	F4 present pre-fix: cosine < 0.5 at $\kappa = 1$ and > 0.999 at $\kappa = 100$
G8	N1 present pre-fix: spurious $\ p - p^*\ > 1$; old-order cost of $p^* > 10\times$ the correct one
G9	N2 present pre-fix (identical J for 3 datasets), absent post-fix
G10	main sweep complete: 3 arms $\times$ 8 seeds $\times$ 16 stages = 384 rows
G11	GP tied with control at $\kappa = 1$ in every seed and strictly better in every non-plateau seed at every $\kappa \geq 2$
G12	paired GP – control CI at $\kappa = 100$ excludes 0
G13	every control row on the plateau returns exactly the seed vector
G14	in-house $S = 10$ loss within 10^{-8} of the adaptive reference at $\kappa \in \{1, 5, 25, 100\}$
G15	other systems: GP better than control in every seed (LV, Lorenz BFGS, Lorenz NM)
G16	every planned sweep job realised
G17	every figure referenced by <code>report.tex</code> exists; results ledger written
G18	Table 2’s medians reproduce from <code>sweeps.csv</code>
G19	the extended landscape animation set is complete: 16 candidate planes screened, 5 animated, 6 two-dimensional and 4 three-dimensional MP4s present and non-trivial
G20	the basin and plateau fractions of Table 8 (and of every animation frame) recompute from the shipped 201×201 grids, all 96 (plane, κ) cells

14 Limitations and honest notes

- **Seeds.** 8 optimiser seeds (6 for noise/sparsity, 4 for iterations/sub-steps/Lorenz, 10 starts per radius for the basin experiment) is enough for paired CIs that exclude zero, not for tail statistics; one seed in eight starts on the plateau and is kept in every median.
- **Nelder–Mead budget.** At 2500 iterations the sweep is converged for FHN (27a) but not for Lorenz (36); the BFGS arms are the recommendation, and their wall times are machine-dependent.
- **The Lipschitz bounds are loose** by orders of magnitude on this attracting system (8–10); the figures support the theory’s *ordering* of single vs multiple shooting, not the size of the constants.
- **Not gated:** F5 (Makie) and F6 (a race) are readings; the concept figures 3 and 7 are schematics and say so.
- **Timing columns** are wall-clock on a shared 16-core machine with up to 16 concurrent processes and are only comparative.
- **Planned vs realised.** 522 planned sweep jobs; the batch was re-sharded twice (resume logic in `04_sweeps.jl`) after two shards stalled on slow L-BFGS jobs; duplicates from deterministic re-runs are removed at merge time (identical results, different wall time). The first-batch sparsity jobs (arm name without γ) were superseded by γ -named arms and are dropped at merge.

Reproduction

See `REPRODUCE.md` for the pinned environment and the ordered commands (Julia stages 00–12, then the Python inventory, tables, figures, animations, gates, number manifest, and `pdflatex` twice). `report.md` is a full-content Markdown rendition of this document generated from the same sources.